

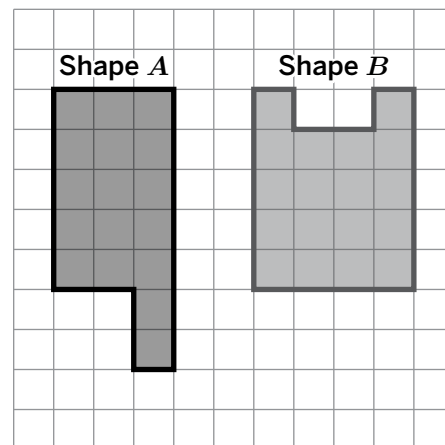
Additional Practice

ACC6.1.01

1. Which shape has a greater area?

Show or explain your thinking.

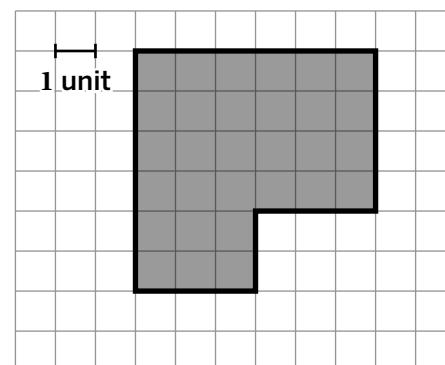
- A. Shape *A*
- B. Shape *B*
- C. They have the same area



Problems 2–4: Here is a new shape.

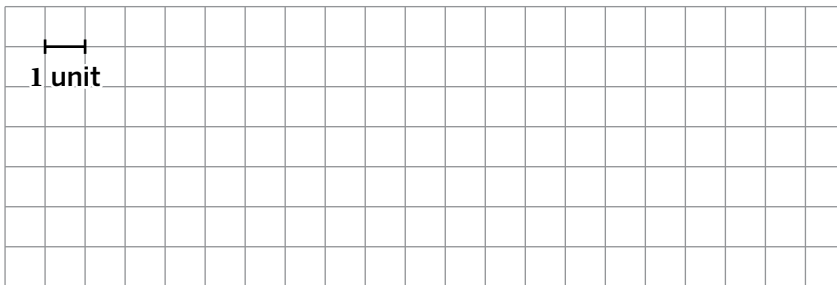
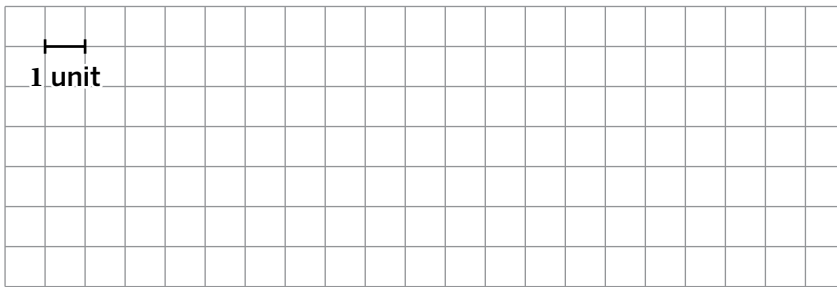
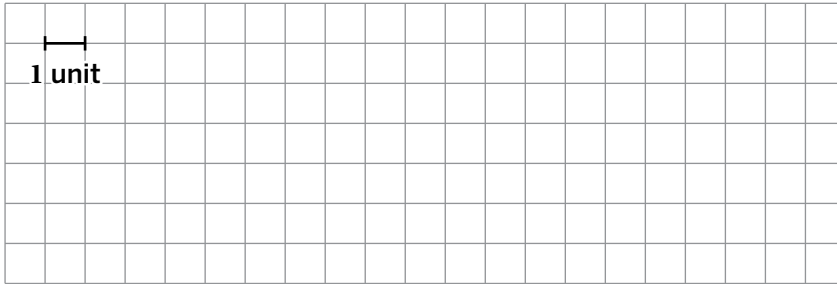
2. Determine the area of the shape.

Show or explain your thinking.



3. Show or describe another way to determine the area of this shape.
4. Show or describe how you would change this shape so it has an area of 20 square units.

5. Draw *three* different quadrilaterals, each with an area of 24 square units.
Each square in this grid has an area of 1 square unit.

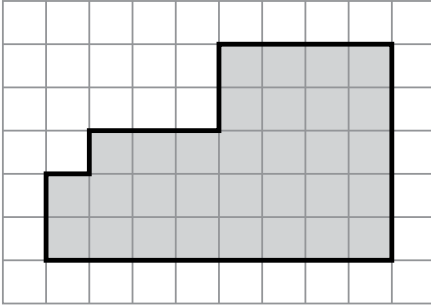


Additional Practice

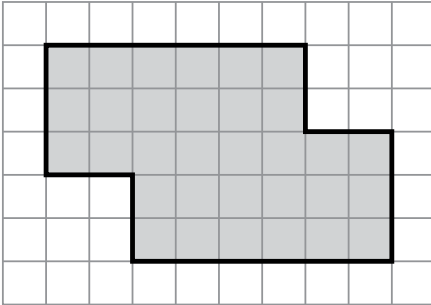
ACC6.1.02

1. Determine the total area of each shaded region.

a

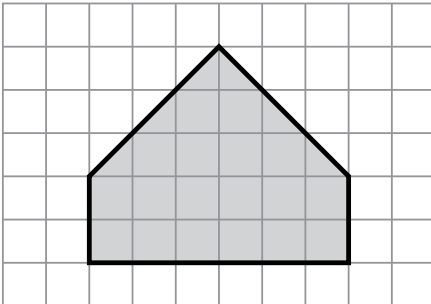


b

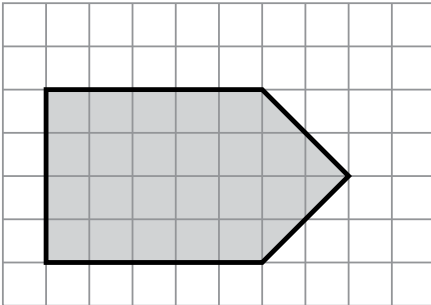


2. Determine the total area of each shaded region.

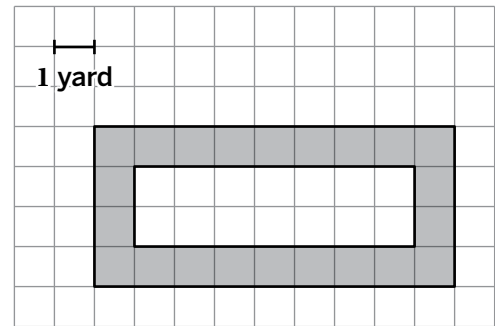
a



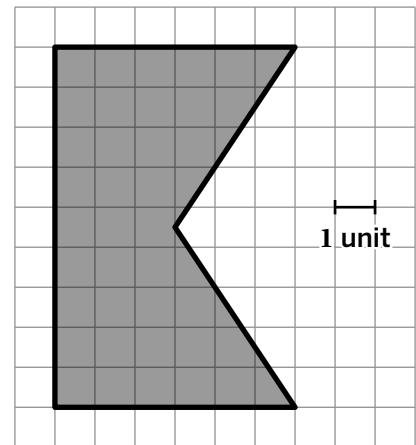
b



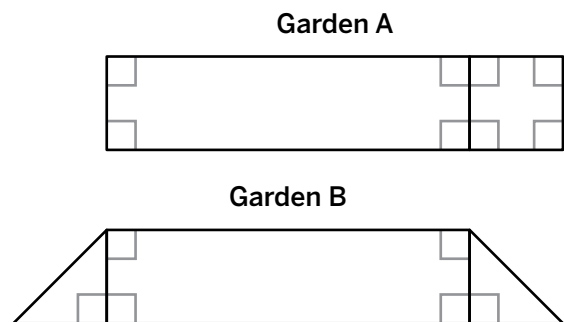
3. Determine the total area of the shaded region.



4. Determine the area of the shape.



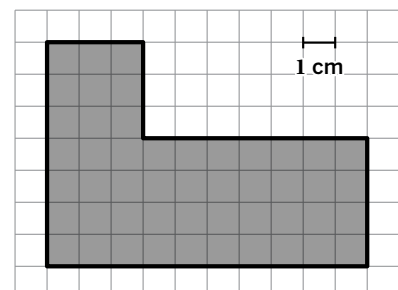
5. Two gardens have very different shapes. Clare said that both gardens have the same area. Is Clare correct? Explain your thinking.



6. Clare calculated the area of this figure. Her work is shown. Is Clare correct? Explain your thinking.

Clare

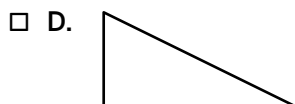
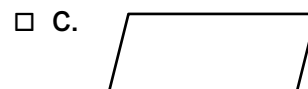
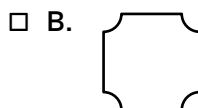
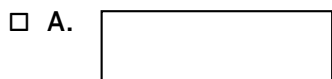
$$\begin{aligned} (10 \times 4) - (3 \times 3) \\ = 40 - 9 \\ = 31 \text{ sq. cm} \end{aligned}$$



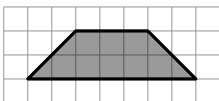
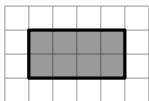
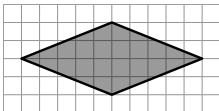
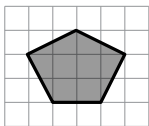
Additional Practice

ACC6.1.03

1. Select *all* of the shapes that appear to be parallelograms.



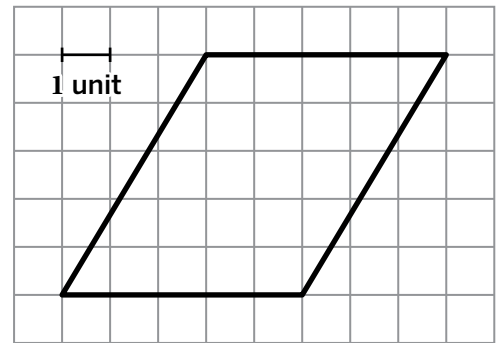
2. Determine whether each figure is a parallelogram. Explain your thinking.

Figure	Parallelogram? (yes/no)	Explanation
a 		
b 		
c 		
d 		

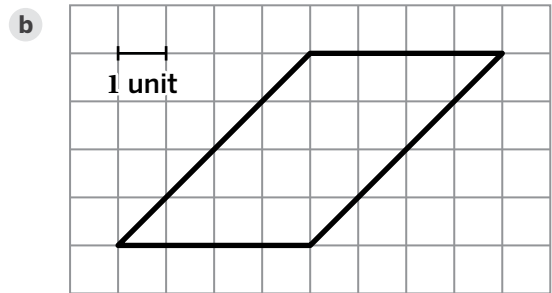
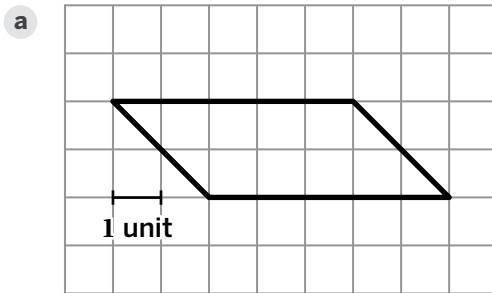
3. Select *all* the true statements.

- ☐ A. A parallelogram can be rearranged to form a rectangle.
- ☐ B. A parallelogram has exactly one pair of parallel sides.
- ☐ C. A parallelogram can have more than four sides.
- ☐ D. In a parallelogram, opposite sides have the same length.
- ☐ E. A square is a parallelogram.

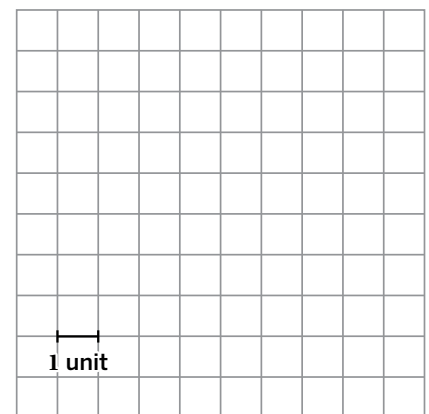
4. Use any strategy to determine the area of the parallelogram.



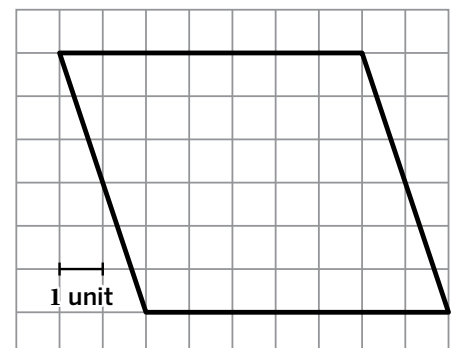
5. Use any strategy to determine the area of each parallelogram.



6. Draw two different parallelograms on the grid, each with an area of 10 square units.



7. Han says he can determine the area of this parallelogram by cutting part of the left side and moving it to the right side to form a rectangle, and that the area of this parallelogram is 36 square units. Is Han correct? Explain your thinking.

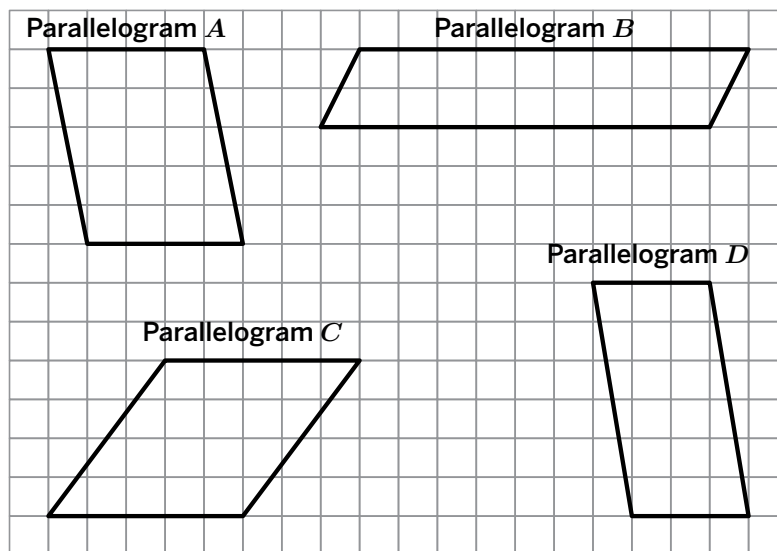


Additional Practice

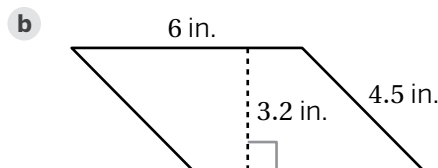
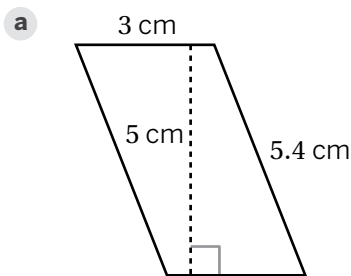
ACC6.1.04

1. Three of these parallelograms have the same area. Which parallelogram has a *different* area than the others?

- A. Parallelogram A
- B. Parallelogram B
- C. Parallelogram C
- D. Parallelogram D



2. Determine the area of each parallelogram.



3. The base lengths, b , and corresponding heights, h , are listed for two different parallelograms. Determine the area of each parallelogram.

a $b = 12.5$ in., $h = 9$ in.

b $b = 6$ cm, $h = 5.5$ cm

4. The base lengths, b , and corresponding heights, h , of four different parallelograms are listed. Which base-height pair represents the parallelogram with the greatest area?

- A. $b = 8$ units, $h = 3.2$ units
- B. $b = 5.6$ units, $h = 4$ units
- C. $b = 10$ units, $h = 2.6$ units
- D. $b = 7.4$ units, $h = 4$ units

5. The base lengths, b , and corresponding areas, A , of four different parallelograms are listed. Determine the height of each parallelogram.

a $b = 12$ units, $A = 84$ square units

b $b = 6$ units, $A = 54$ square units

c $b = 5.5$ units, $A = 33$ square units

d $b = 8.2$ units, $A = 28.7$ square units

6. The heights, h , and corresponding areas, A , of four different parallelograms are listed. Determine the length of the base of each parallelogram.

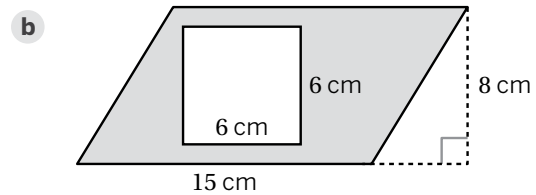
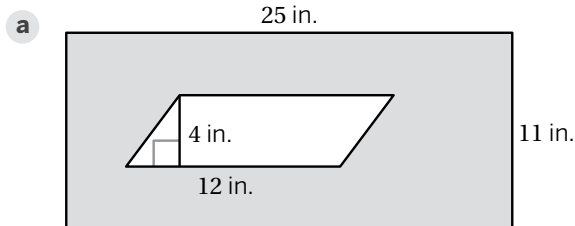
a $h = 4$ units, $A = 16$ square units

b $h = 7$ units, $A = 35$ square units

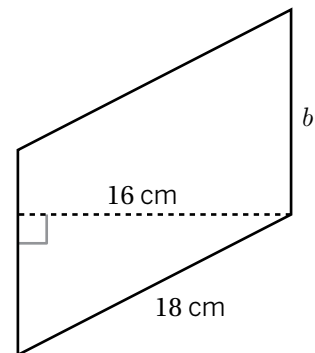
c $h = 2.5$ units, $A = 10$ square units

d $h = 8$ units, $A = 28$ square units

7. Determine the area of the shaded region in each figure.



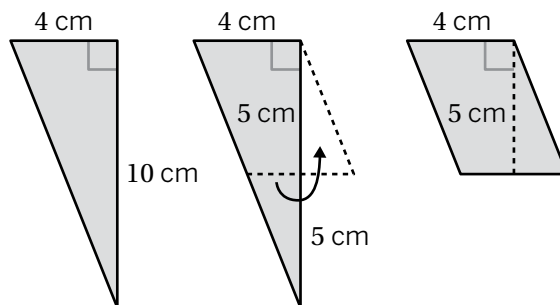
8. The parallelogram shown has an area of 144 square centimeters. Mai claims the length of the base, b , is 8 centimeters. Is Mai correct? Explain your thinking.



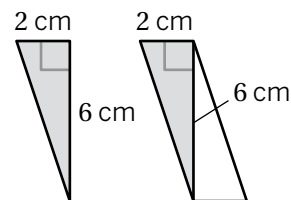
Additional Practice

ACC6.1.05

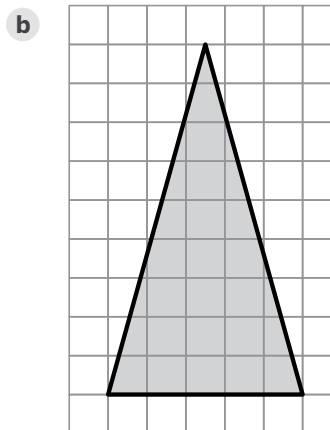
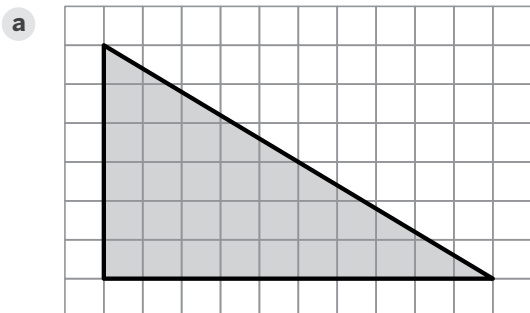
- To determine the area of this triangle, Jada decomposed and rearranged it into a parallelogram. Explain how Jada could use her parallelogram to determine the area of the triangle.



- To determine the area of this triangle, Lin used two identical copies of the triangle to create a parallelogram. Explain how Lin could use her parallelogram to determine the area of the triangle.

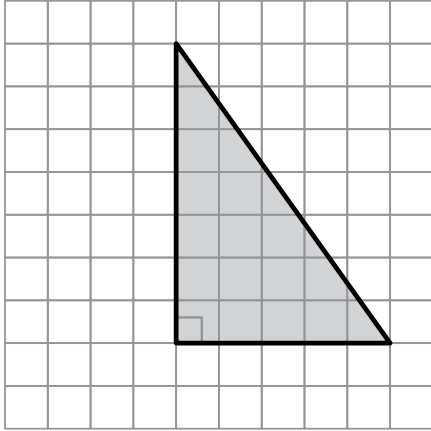


- Determine the area of each triangle. Each small square in the grid has an area of 1 square unit.

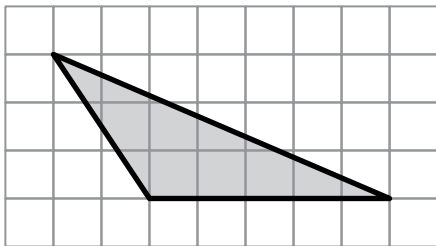


Problems 4–6: Determine the area of each triangle. Show or explain your thinking. Each small square in the grid has an area of 1 square unit.

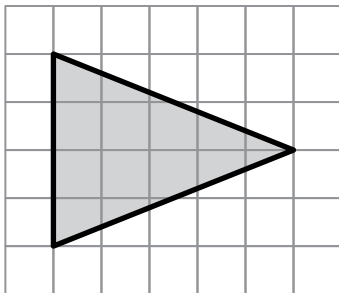
4.



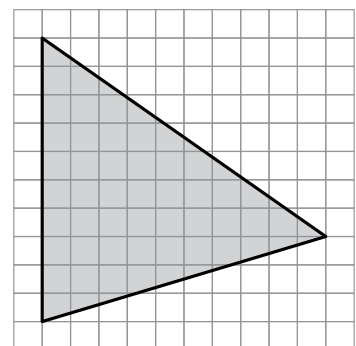
5.



6.



7. Shawn says the area of this triangle is 100 square units. Is Shawn correct? Explain your thinking.



Additional Practice

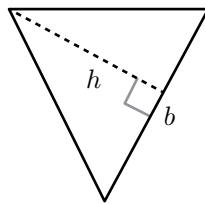
ACC6.1.06

1. Select *all* the statements that are true about triangles.

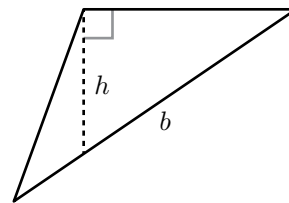
- ☐ A. Any side of a triangle can be a base.
- ☐ B. The height must be drawn inside the triangle.
- ☐ C. A height that corresponds to the base of a triangle is always perpendicular to the base.
- ☐ D. A height that corresponds to the base of a triangle is always drawn at a right angle to the base.
- ☐ E. The height is sometimes drawn outside the triangle.

2. Select *all* of the triangles that have a correct height labeled for the given base.

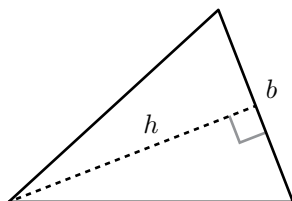
☐ A. **Triangle A**



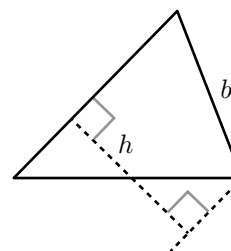
☐ B. **Triangle B**



☐ C. **Triangle C**

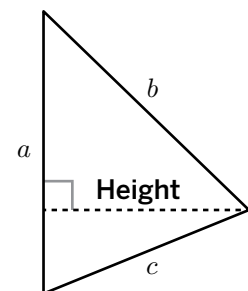


☐ D. **Triangle D**



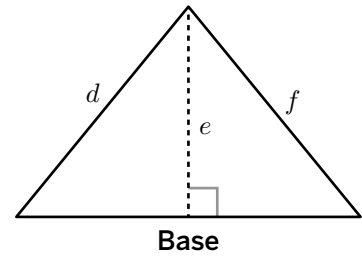
3. Which is a matching base for the indicated height of the triangle?

- A. Side a
- B. Side b
- C. Side c



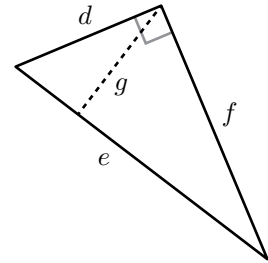
4. Which is a height that matches the given base of the triangle?

- A. Side d
- B. Length e
- C. Side f



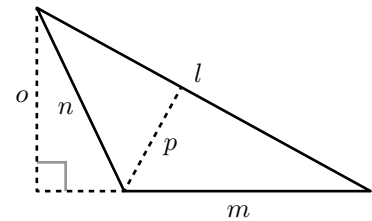
5. Name a matching height for each indicated base.

- a Side d
- b Side e
- c Side f

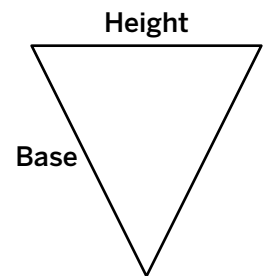


6. Name a matching height for each indicated base.

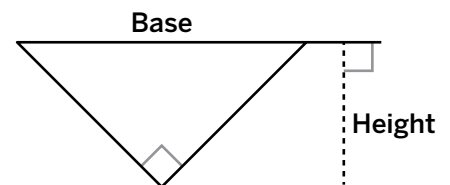
- a Side l
- b Side m



7. Tyler labeled the base and height of the triangle. Are Tyler's labels correct? Explain your thinking.



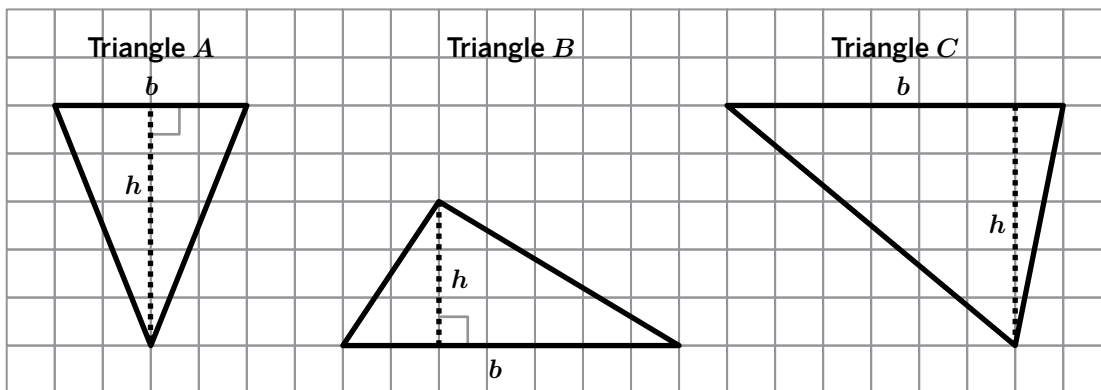
8. Elena labeled the base and height of the triangle. Are Elena's labels correct? Explain your thinking.



Additional Practice

ACC6.1.07

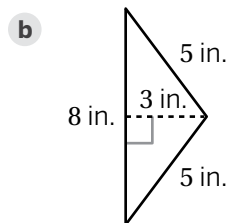
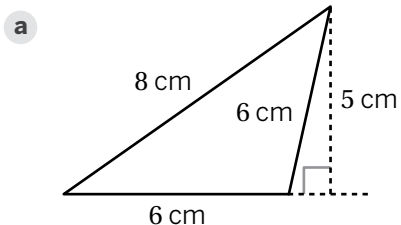
1. For each triangle, a base, b , and its corresponding height, h , are labeled. Each small square in the grid has an area of 1 square unit.



Complete the table.

	Base (units)	Height (units)	Area (square units)
Triangle A			
Triangle B			
Triangle C			
Any Triangle	b	h	

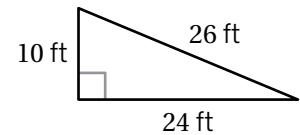
2. Determine the area of each triangle.



3. Use the given bases and heights to determine the area of each triangle in the table.

	Base (units)	Height (units)	Area (square units)
Triangle A	8	6	
Triangle B	16	5	
Triangle C	3	3	
Triangle D	10.5	5	

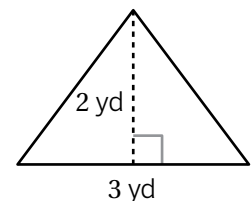
4. Determine the area of the triangle. Show your thinking.



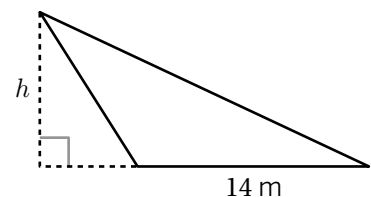
5. A triangle has a base with a length of 24.8 centimeters and a height of 16 centimeters. Determine its area.

6. A triangle has a base with a length of 7 inches and a height of 12 inches. Determine its area.

7. A herb garden in Bard's backyard is shaped like a triangle with the dimensions shown. Determine the area of the herb garden.



8. A triangle has a base of 14 meters and an area of 70 square meters. What is the height of this triangle? Explain your thinking.

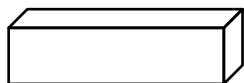


Additional Practice

ACC6.1.08

1. Select *all* the polygons.

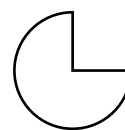
☐ A.



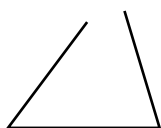
☐ B.



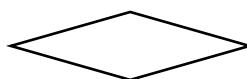
☐ C.



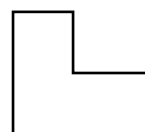
☐ D.



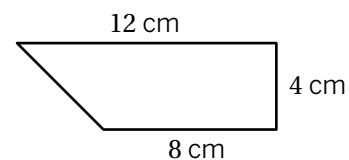
☐ E.



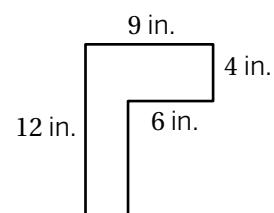
☐ F.



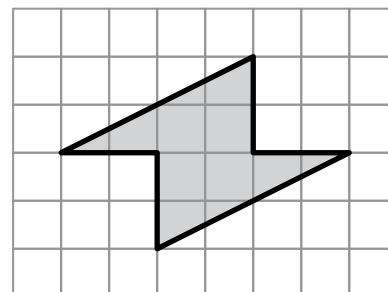
2. Use any strategy to determine the area of the polygon.
Explain your thinking.



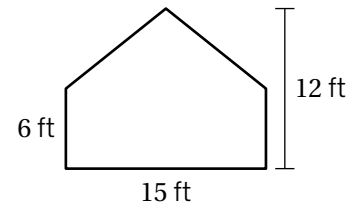
3. Use any strategy to determine the area of the polygon.
Explain your thinking.



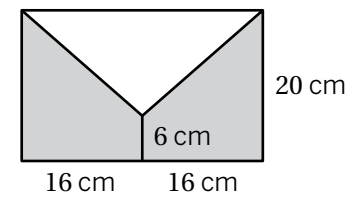
4. Use any strategy to determine the area of the polygon.
Explain your thinking.



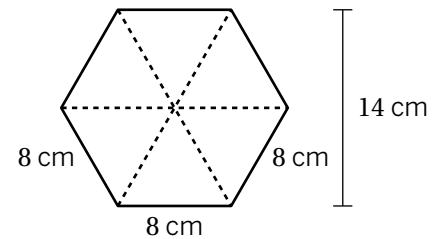
5. Use any strategy to determine the area of the polygon.
Explain your thinking.



6. Determine the area of the shaded region.
Explain your thinking.



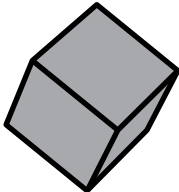
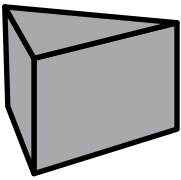
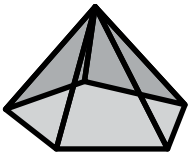
7. The hexagon has a side length of 8 centimeters. Diego says: *The area is 28 square centimeters because each triangle has a base of 8 centimeters and a height of 7 centimeters. If I multiply 8 by 7 and divide the product by 2, I get 28 square centimeters.* Is Diego correct? Explain your thinking.



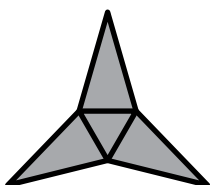
Additional Practice

ACC6.1.09

Problems 1–5: Here are some polyhedra.

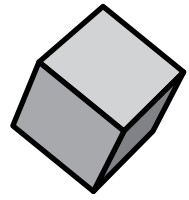
Figure A	Figure B	Figure C	Figure D	Figure E
				

- Which polyhedra are prisms?
- Which polyhedra are pyramids?
- What type of polyhedron is figure *D*?
- What type of polyhedron is figure *C*?
- Which of these polyhedra could be created from this net?

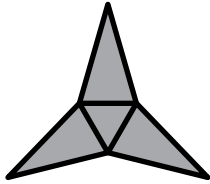


6. Here is a polyhedron.

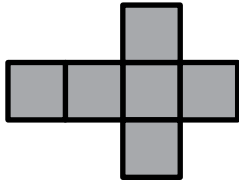
Which net could create this polyhedron?



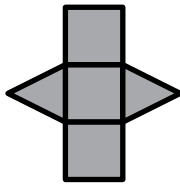
A.



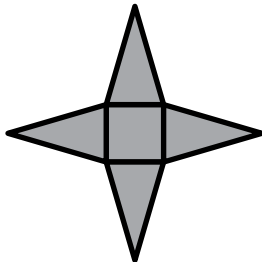
B.



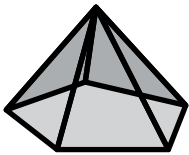
C.



D.



7. What polygons make up the faces of this three-dimensional figure?



Additional Practice

ACC6.1.10

1. Select *all* the units that can be used to describe surface area.

☐ A. Square inches

☐ B. Meters

☐ C. Centimeters

☐ D. Cubic feet

☐ E. Square meters

☐ F. Cubic inches

2. Select *all* the units that can be used to describe volume.

☐ A. Square inches

☐ B. Meters

☐ C. Centimeters

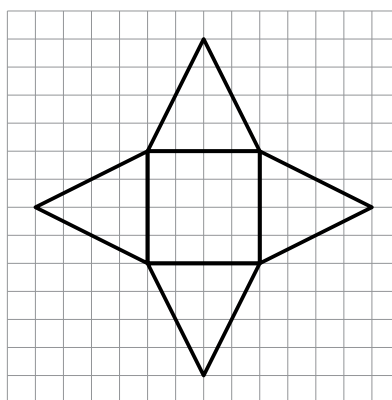
☐ D. Cubic feet

☐ E. Square meters

☐ F. Cubic inches

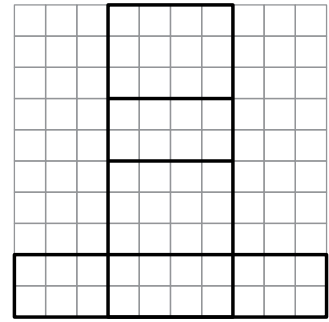
3. What is the difference between the units used to measure surface area and the units used to measure volume? Look back at your work on Problems 1–2 if it is helpful.

4. Here is a net.

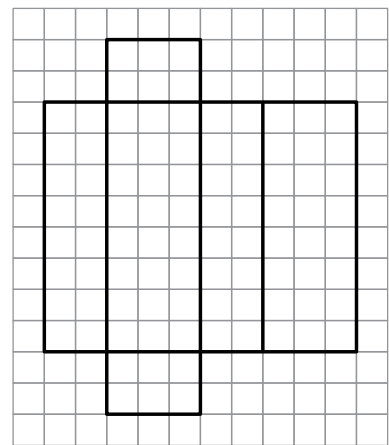


Determine the surface area, in square units, of the polyhedron that could be created using this net.

5. Determine the surface area of the rectangular prism that could be created from this net.

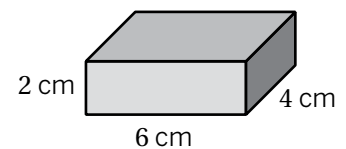


6. Determine the surface area, in square units, of the rectangular prism that would be formed by this net. Show or explain your thinking.



7. Here is a rectangular prism.

- a Use the grid to draw a net for the prism. The length of one grid square is 1 centimeter.

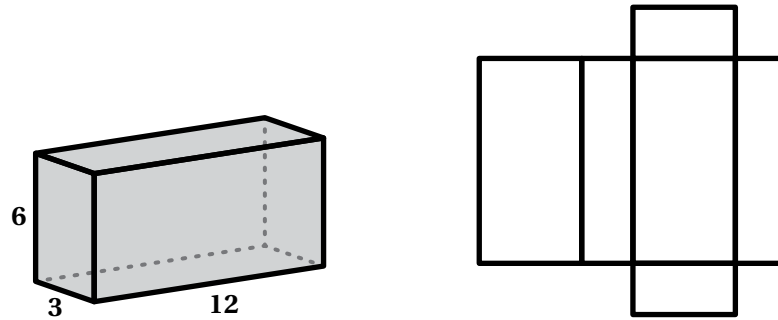


- b Determine the surface area of the prism.

Additional Practice

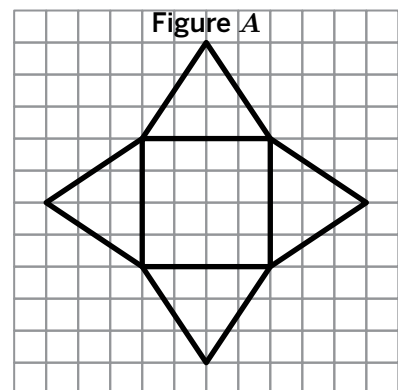
ACC6.1.11

Problems 1–2: Here is a rectangular prism and its matching net.



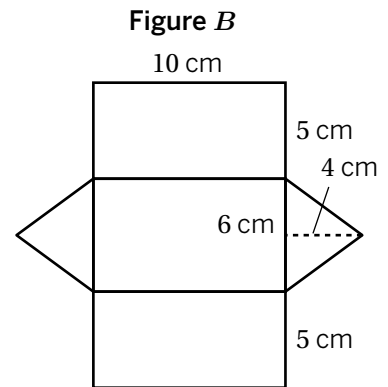
1. Use the rectangular prism to label all the lengths in this net.
2. Use the net to calculate the surface area in square units. Show or explain your thinking.

- 3.** The net for figure *A* is shown.
- Name the type of polyhedron that the net would form when assembled.
 - Determine the surface area of the polyhedron.

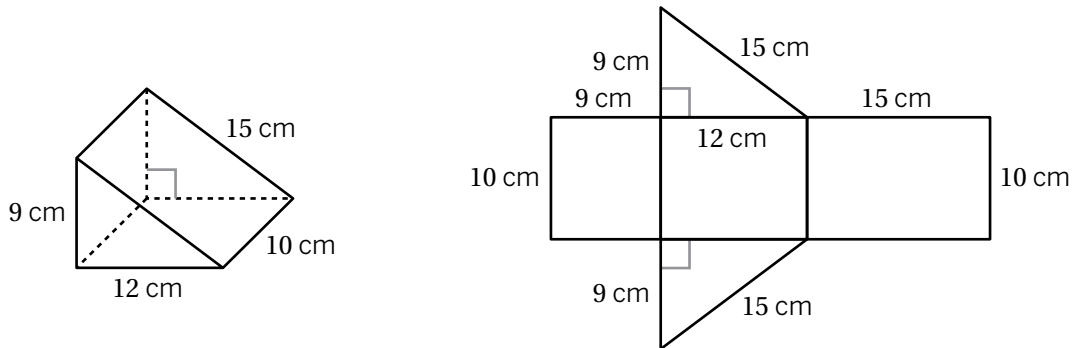


4. The net for figure *B* is shown.

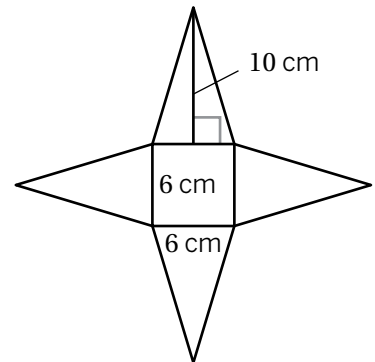
- a Name the type of polyhedron that the net would form when assembled.
- b Determine the surface area of the polyhedron.



5. Here is a triangular prism and its matching net. Determine the surface area.



6. Bard claims the surface area of this square pyramid is 192 square centimeters. Is Bard correct? Explain your thinking.



Additional Practice

ACC6.2.01

1. Here are sets of shapes. Complete each statement.

a **Triangles**  

Circles  

There are _____ triangles for every _____ circles.

b **Squares**  

Diamonds  

The ratio of diamonds to squares is _____ to _____.

2. Select *all* the statements that correctly describe the relationship between the almonds and pretzels in this snack mix.

Almonds  
Pretzels  

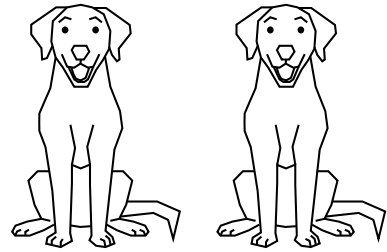
- ☐ A. The ratio of almonds to pretzels is 3 to 2.
- ☐ B. For every 2 pretzels, there are 3 almonds.
- ☐ C. The ratio of almonds to pretzels is 2 : 3.
- ☐ D. For every 3 almonds, there are 2 pretzels.
- ☐ E. For every 6 pretzels, there are 2 almonds.
- ☐ F. The ratio of almonds to pretzels is 4 : 6.

3. A recipe for banana bread uses 4 bananas, 2 eggs, and 1 cup of sugar.

Bananas  
Eggs  
Sugar (cups)  

- a The ratio of eggs to sugar is _____ : _____.
- b The ratio of sugar to bananas is _____ to _____.
- c There are _____ bananas for every 1 egg.

Problems 4–5: Here is a picture of two dogs.



- 4.** Which diagram represents the ratio relationship between the number of eyes, tails, and feet?

A. Eyes  

Tails        

Feet    

B. Eyes  

Tails  

Feet  

C. Eyes  

Tails  

Feet  

D. Eyes  

Tails    

Feet        

- 5.** Complete each statement.

- The ratio of eyes to tails is :
- The ratio of feet to tails is to
- There are eyes for every tail.
- There are feet for every tail.

- 6.** The diagram represents the number of percussion players and woodwind players in a school band.

Percussion 

Woodwind 

Clare says the ratio of woodwind players to percussion players is 2 : 6. Andre says the ratio of woodwind players to percussion players is 1 : 3. Who is correct? Explain your thinking.

7. The diagram represents the number of cups of water and tablespoons of loose tea to make iced tea. Mai says the ratio of water to tea is 4 : 2. What is another ratio that Mai could write for the ratio of water to tea? Explain your thinking.

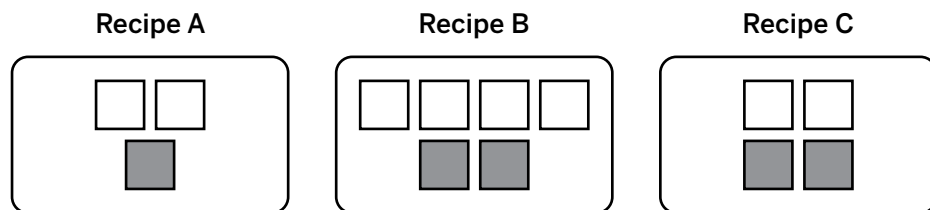
Water (cups) 

Tea (tbsp) 

Additional Practice

ACC6.2.02

Problems 1–4: Here are three recipes for green dye using yellow and blue food coloring.

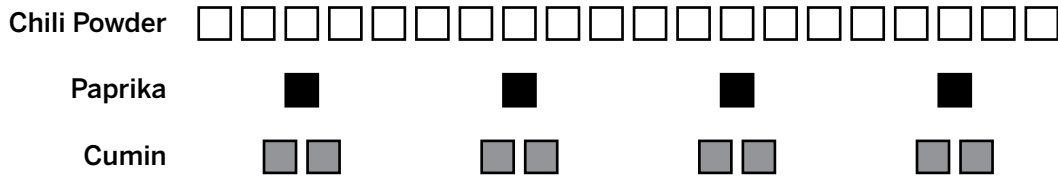


Key:

□ = 1 drop yellow food coloring ■ = 1 drop blue food coloring

1. Use the diagrams to complete each pair of statements.
 - a Recipe A uses _____ drop(s) of yellow food coloring and _____ drop(s) of blue food coloring.
The ratio of drops of yellow to blue in Recipe A is _____.
 - b Recipe B uses _____ drop(s) of yellow food coloring and _____ drop(s) of blue food coloring.
The ratio of drops of yellow to blue in Recipe B is _____.
 - c Recipe C uses _____ drop(s) of yellow food coloring and _____ drop(s) of blue food coloring.
The ratio of drops of yellow to blue in Recipe C is _____.
2. Which statement is *true* about the ratios of the recipes?
 - A. The quantities in Recipe A are double those of Recipe B.
 - B. The quantities in Recipe B are double those of Recipe C.
 - C. The quantities in Recipe C are double those of Recipe A.
 - D. The quantities in Recipe B are double those of Recipe A.
3. To darken the green dye, you can add more drops of blue. Which recipe has the darkest shade of green dye? Explain your thinking.
4. What could be added to Recipe B so that the color will be the same as Recipe C?

5. A recipe for 1 batch of taco seasoning says, “Mix 5 parts chili powder, 1 part paprika, and 2 parts cumin.” How many batches are represented by the diagram? Explain your thinking.



6. Kiran makes banana oatmeal pancakes by mixing 2 mashed bananas with 3 cups of oatmeal (along with some other ingredients).

- a Draw a diagram that represents doubling the recipe for banana oatmeal pancakes.

Bananas

Oatmeal (cups)

- b Draw a diagram that represents tripling the recipe for banana oatmeal pancakes.

Bananas

Oatmeal (cups)

7. In a recipe for sparkling orange juice, the ratio of cups of orange juice to cups of sparkling water is 2 : 1.

- a Write the ratio of cups of orange juice to cups of sparkling water that represents a double batch.

- b Write the ratio of cups of orange juice to cups of sparkling water that represents a quadruple batch.

8. When Elena makes one batch of hummingbird food, she mixes 4 cups of water with 1 cup of sugar, which is enough for one hummingbird feeder.

- a If Elena has 5 hummingbird feeders in her yard, what is the ratio of cups of water to cups of sugar that she would use to fill all 5 feeders?

- b Elena says that if she has 3 hummingbird feeders in her yard, she would need 2 cups of sugar and 8 cups of water. Is Elena correct? Explain your thinking.

Additional Practice

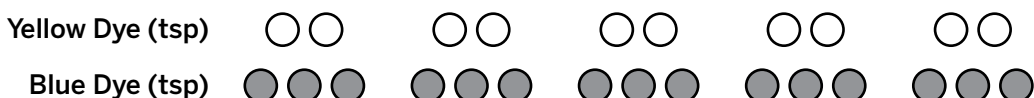
ACC6.2.03

1. The diagram shows a mixture of black paint and white paint needed for 3 batches of a gray paint.



What is the ratio of black paint to white paint needed for 1 batch?

2. The diagram shows a mixture of yellow dye and blue dye needed for 5 batches of a green dye.



What is the ratio of blue dye to yellow dye needed for 1 batch?

3. Bard makes pink paint by mixing 16 tablespoons of white paint and 4 tablespoons of red paint. Select *all* of the mixtures that produce the same pink paint as Bard's mixture, but in a smaller amount.

- ☐ A. Mix 15 tablespoons of white paint and 3 tablespoons of red paint.
- ☐ B. For every tablespoon of red paint, mix 4 tablespoons of white paint.
- ☐ C. Mix tablespoons of red paint and white paint in the ratio 4 : 1.
- ☐ D. For every 2 tablespoons of red paint, mix 8 tablespoons of white paint.
- ☐ E. Mix 8 tablespoons of white paint with 4 tablespoons of red paint.

4. Lin makes sparkling lemonade by mixing 12 cups of lemonade and 16 cups of sparkling water.

- a What is the ratio of sparkling water to lemonade?
- b If Lin uses 6 cups of lemonade, how many cups of sparkling water should she use so the taste remains the same?
- c If Lin uses 4 cups of sparkling water, how many cups of lemonade should she use so the taste remains the same?

5. To make 1 batch of pale yellow paint, Diego mixes 3 cups of yellow paint with 1 gallon of white paint. Select *all* the mixtures that are a darker tint of yellow than the pale yellow.

- ☐ A. Add more white paint to the mixture.
- ☐ B. Mix 5 cups of yellow paint with 1 gallon of white paint.
- ☐ C. Mix 2 gallons of white paint with 3 cups of yellow paint.
- ☐ D. Add more yellow paint to the mixture.
- ☐ E. Mix 3 cups of yellow paint with $\frac{1}{2}$ gallon of white paint.

6. When dyeing yarn, use 750 milliliters of water for every 25 grams of yarn fibers. Complete the table.

Water (mL)	Yarn Fibers (g)
750	25
150	
	1

7. To make a large batch of pancake mix, the directions say to use 40 cups of water and 28 cups of pancake mix.

- a Diego only needs half the amount of pancake mix. What ratio would represent half of the recipe?
- b Lin wants to use 8 cups of water and 7 cups of pancake mix. Is her ratio equivalent to the ratio in the directions? Explain your thinking.

8. Orange paint can be made by mixing 35 teaspoons of red paint and 14 teaspoons of yellow paint. Kiran and Priya each attempted to make a smaller amount of the same orange paint color.

Kiran		Priya	
Red (tsp)	☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐	Red (tsp)	☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐
Yellow (tsp)	☐ ☐	Yellow (tsp)	☐ ☐ ☐ ☐ ☐ ☐

Does either person's color mixture make the same orange color as the original orange paint? Explain your thinking.

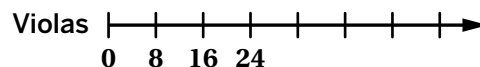
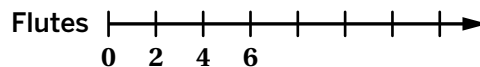
Additional Practice

ACC6.2.04

1. In an orchestra, the ratio of flutes to violas is 2 : 8.

List two other combinations of flutes and violas for a different orchestra with the same ratio of instruments.

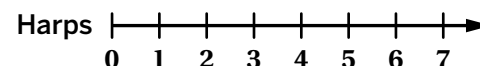
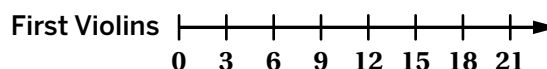
Flutes	Violas



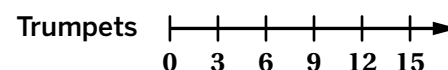
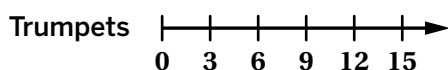
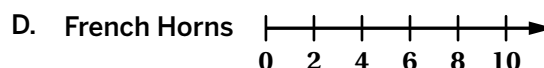
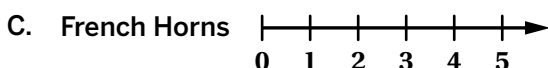
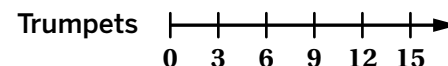
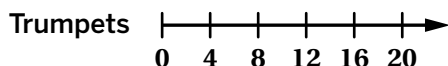
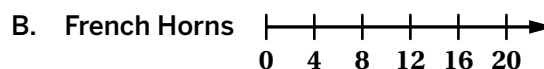
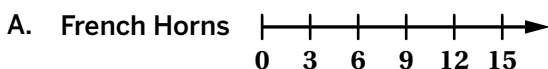
2. In an orchestra, the ratio of first violins to harps is 3 : 1.

List two other combinations of first violins and harps for a *larger* orchestra with the same ratio of instruments.

First Violins	Harps

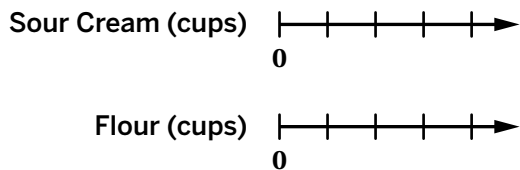


3. In an orchestra, the ratio of French horns to trumpets is 4 : 3.
Which double number line correctly shows this ratio?



Problems 4–7: Here are the ingredients of a recipe for one batch of blueberry muffins.

4. Complete the double number line representing the amounts of sour cream and flour.

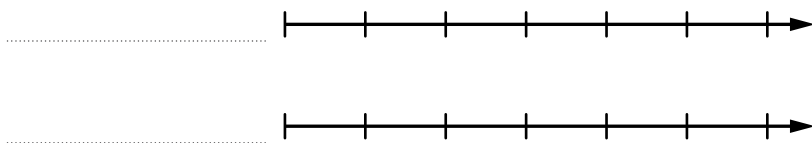


Ingredient	Amount
Blueberries	$2\frac{1}{4}$ cups
Sour cream	1 cup
Salt	$\frac{1}{2}$ tsp
Flour	2 cups
Baking powder	3 tsp
Sugar	1 cup

5. Bard plans on making several batches of blueberry muffins.

- a If Bard uses $2\frac{1}{2}$ teaspoons of salt, how many batches could be made?
- b If Bard uses 12 teaspoons of baking powder, how many batches could be made?

6. Draw a double number line to represent the amounts of sugar and baking powder in the recipe.



7. Diego wants to make 3 batches of blueberry muffins. He says he will need $6\frac{1}{2}$ cups of blueberries, 6 cups of flour, and 3 teaspoons of salt. Is Diego correct? Explain your thinking.

Additional Practice

ACC6.2.07

1. Determine *all* the factors of each number.

a 16

b 25

c 100

d 37

2. For the numbers 20 and 24, determine:

a The factors of 20.

b The factors of 24.

c The common factors of 20 and 24.

d The greatest common factor (GCF) of 20 and 24.

3. For the numbers 18 and 27, determine:

a The factors of 18.

b The factors of 27.

c The common factors of 18 and 27.

d The greatest common factor (GCF) of 18 and 27.

4. Determine the greatest common factor for each pair of numbers.

a 33 and 15

b 8 and 32

c 45 and 54

d 35 and 21

5. Consider the numbers 42 and 96.

- a** How many common factors do 42 and 96 have? Explain your thinking.
- b** What is the greatest common factor of 42 and 96? Explain your thinking.

6. A teacher is making gift bags. Each bag will have pencils and erasers. The teacher has 40 pencils and 60 erasers to use. Each bag will have the same number of each item, with no items left over. Select *all* of the possible numbers of pencils and erasers in the gift bags.

- ☐ **A.** 10 bags with 4 pencils and 6 erasers
- ☐ **B.** 4 bags with 6 pencils and 4 erasers
- ☐ **C.** 20 bags with 2 pencils and 6 erasers
- ☐ **D.** 2 bags with 20 pencils and 30 erasers
- ☐ **E.** 5 bags with 8 pencils and 12 erasers

7. A middle school band has 72 sixth grade students and 54 seventh grade students. The band director wants to make groups of performers, with the same combination of sixth grade and seventh grade students in each group.

- a** What is the greatest number of groups that could be formed? Explain your thinking.
- b** Using your answer from Problem 7a, how many sixth graders would be in each group?
- c** Using your answer from Problem 7a, how many seventh graders would be in each group?

8. Jada, Shawn, and Noah were all asked to determine the greatest common factor of 16, 72, and 80. Their responses are shown.

- Jada says that the greatest common factor is 16.
- Shawn says that the greatest common factor is 8.
- Noah says that the greatest common factor is 9.

Who is correct? Explain your thinking.

Additional Practice

ACC6.2.08

1. Priya has 2 white marbles and 4 blue marbles. Lin has 4 white marbles and 16 blue marbles. Who has the greater ratio of white marbles to blue marbles? Explain your thinking.
 - A. Priya
 - B. Lin
 - C. They have the same ratio of white to blue marbles.

2. Kiran shops at a bulk store where he can purchase dry goods by the ounce.

- a Complete the table.

Item	Price (\$)	Number of Ounces	Price per Ounce (\$)
Banana chips	9	36	
Macadamia nuts	16	8	
Dried mango	8	8	
Cinnamon pecans	18	24	
Pistachios	3.75	5	

- b Which item has the lowest price per ounce? Explain your thinking.

3. Here are 3 possible mixtures to make pink paint.
 - Mixture A is made with 18 teaspoons of white paint and 12 teaspoons of red paint.
 - Mixture B is made with 6 teaspoons of white paint and 5 teaspoons of red paint.
 - Mixture C is made with 12 teaspoons of white paint and 10 teaspoons of red paint.

Which mixture is the *lightest* shade of pink paint? Explain your thinking.

- 4.** Noah's team won 10 games and lost 5 games. Tyler's team won 12 games and lost 4 games. Whose team had the greater ratio of wins to losses? Explain your thinking.

A. Noah's team **B.** Tyler's team **C.** Both teams have the same ratio of wins to losses.

- 5.** Movie tickets cost different amounts depending on the day and time of the show.

- Bard paid \$39 for 6 tickets.
- Andre paid \$21.75 for 3 tickets.
- Mai paid \$31.25 for 5 tickets.

Who paid the lowest price per ticket? Explain your thinking.

A. Bard **B.** Andre **C.** Mai

- 6.** Diego swam 200 meters in 58 seconds. Han swam 300 meters in 87 seconds. Both swam at a constant speed. Who swam at a faster constant speed? Explain your thinking.

A. Diego **B.** Han **C.** They swam at the same speed.

- 7.** Clare ran 3 miles in 27 minutes. Jada ran 5 miles in 40 minutes. Both ran at a constant speed. Did they run at the *same* constant speed? Explain your thinking.

- 8.** Shawn paid \$11.25 for 2.5 pounds of chicken. Lin paid \$25.50 for 6 pounds of chicken. Lin says she paid less per pound of chicken than Shawn. Is Lin correct? Explain your thinking.

Additional Practice

ACC6.2.09

1. To make a fruit smoothie, Andre uses $\frac{1}{2}$ cup of almond milk and $1\frac{1}{2}$ cups of frozen berries. The table shows equivalent ratios of almond milk and frozen berries.

Almond Milk (cups)	Frozen Berries (cups)
$\frac{1}{2}$	$1\frac{1}{2}$
1	3
$1\frac{1}{2}$	$4\frac{1}{2}$
2	6

Select *all* of the true statements about the ratio of almond milk to frozen berries.

- ☐ A. For every 3 cups of berries, there is 1 cup of almond milk.
- ☐ B. For every 2 cups of berries, there are 6 cups of almond milk.
- ☐ C. For every $1\frac{1}{2}$ cups of berries, there is 1 cup of almond milk.
- ☐ D. For every 1 cup of almond milk, there are 3 cups of berries.
- ☐ E. For every $\frac{1}{2}$ cup of almond milk, there are $1\frac{1}{2}$ cups of berries.
2. A particular pink paint is made by mixing 3 parts of red paint with 9 parts of white paint. Complete the table with the amount of red paint and white paint needed to make different amounts of the same shade of pink paint.

Red Paint	White Paint
1	
$\frac{1}{2}$	
	12
	6

- 3.** In a recipe for waffles, there are $2\frac{1}{4}$ cups of flour for every 2 cups of milk. A family is making several batches of waffles. Determine how much of each ingredient the family will need. Use a table if it helps with your thinking.

- a** How many cups of milk are needed to make 3 batches of waffles?

- b** How many cups of flour are needed to make 3 batches of waffles?

- c** How many cups of milk are needed to make 5 batches of waffles?

- d** How many cups of flour are needed to make 6 batches of waffles?

- 4.** A car travels at a constant speed and its distance traveled in 1, 2, and 3 hours is shown in the table. How far does the car travel in 8 hours? Show or explain your thinking.

Time (hours)	Distance (miles)
1	65
2	130
3	195

- 5.** Complete the table by writing three equivalent ratios to 24 : 15.

x	y
24	15

- 6.** A recipe for a batch of bran muffins calls for $\frac{3}{4}$ cups of brown sugar and 3 cups of bran cereal. Andre says that, if he triples the recipe, he will need $3\frac{3}{4}$ cups of brown sugar and 9 cups of bran cereal. Is Andre correct? Show or explain your thinking.

Additional Practice

ACC6.2.10

1. Determine the missing number in the ratio table.

Number of Knives	Number of Spoons
7	12
	48

2. A recipe calls for 2 pounds of chicken and $3\frac{1}{2}$ cups of tomato sauce. Andre makes 3 batches of the recipe. What amount of ingredients are needed for 3 batches?
- A. 5 pounds of chicken, 9 cups of tomato sauce
 - B. 5 pounds of chicken, $10\frac{1}{2}$ cups of tomato sauce
 - C. 6 pounds of chicken, 9 cups of tomato sauce
 - D. 6 pounds of chicken, $10\frac{1}{2}$ cups of tomato sauce
3. A chef is using vinegar to pickle onions. The restaurant supply store sells 2 gallons of vinegar for \$4.88. Which ratios correctly represent the price of the vinegar? Select *all* that apply.
- ☐ A. 7 gallons to \$14.64
 - ☐ B. 14 gallons to \$34.16
 - ☐ C. 1 gallon to \$2.44
 - ☐ D. 20 gallons to \$43.92
 - ☐ E. 5 gallons to \$12.20
4. A full-grown elephant drinks about 1,330 liters of water each week. Complete each ratio based on this information.
- a _____ liters to 1 day
 - b 570 liters to _____ days
 - c _____ liters to 3 weeks
 - d 2,280 liters to _____ days
 - e _____ liters to 12 weeks

- 5.** A caterer needs to buy 38 pounds of potatoes for an event. It costs \$6 for 5 pounds of potatoes.

- a** Write a ratio for the price of the potatoes per pound.
- b** If all potatoes are sold at this rate, how much will the caterer pay for the potatoes they need? Explain your thinking.

- 6.** A caterer also needs to buy 130 rolls for the event. A bakery sells 12 rolls for \$4.80.

- a** Write a ratio to describe the cost of the rolls.
- b** If all rolls are sold at the same rate, how much will the caterer pay for the rolls they need? Explain your thinking.

- 7.** The directions for a shade of green paint say to mix 7 milliliters of yellow paint with 3 milliliters of blue paint. Han has 42 milliliters of yellow paint and 25 milliliters of blue paint. If Han wants to use all of the yellow paint, how much blue paint, if any, will he have left? Show or explain your thinking. Use the ratio table it it helps you show your thinking.

Yellow (mL)	Blue (mL)

- 8.** Mai is reading a 55-page book. She read the first 35 pages in 30 minutes. If she continues to read at the same rate, will she be able to complete this book in less than 1 hour? Show or explain your thinking.

Additional Practice

ACC6.2.11

Problems 1–2: The ratio of cats to dogs at a boarding facility is 3 : 8. There are 33 total cats and dogs staying at the boarding facility.

1. How many cats are at the boarding facility?
2. How many dogs are at the boarding facility?

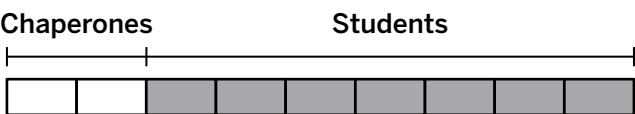
Problems 3–4: Here is a tape diagram representing the ratio of yellow paint to blue paint in a mixture of green paint.



3. What is the ratio of yellow paint to blue paint?
4. Complete the table to show the amount of yellow and blue paint needed to make each quantity of green paint.

Green (gal)	Yellow (gal)	Blue (gal)
21		
35		

Problems 5–6: Here is a tape diagram representing the ratio of students to chaperones on a school trip.



5. What is the ratio of chaperones to students?
6. Complete the table to show the number of chaperones and students for each amount of people on the school trip.

Number of People	Chaperones	Students
18		
90		

7. Jada has a bag of 100 marbles. The ratio of red marbles to blue marbles is 4 : 1. How many blue marbles does she have? Show or explain your thinking.
8. A veterinarian examined 54 pets at her clinic in one day. The ratio of dogs to other pets was 8 : 1. How many dogs were examined? Show or explain your thinking.

Additional Practice

ACC6.2.12

Problems 1–2: Pasta is made from 2 parts flour and 5 parts water.

1. How much flour is needed to make 28 ounces of pasta?
2. How much water is needed to make 28 ounces of pasta?

Problems 3–4: A smoothie is made from 6 parts fruit and 2 parts milk.

3. How much fruit is needed to make 24 ounces of smoothie?
4. How much milk is needed to make 24 ounces of smoothie?

Problems 5–6: One batch of purple paint uses 6 teaspoons of blue paint for every 9 teaspoons of red paint.

Blue Paint (tsp)	Red Paint (tsp)	Purple Paint (tsp)
		45
		10

5. Complete the table to show the amount of blue and red paint needed to make each amount of purple paint.
6. Shawn made 5 teaspoons of purple by combining blue paint and red paint. Shawn used 3 teaspoons of blue paint. Is Shawn using the same recipe? Explain your thinking.

- 7.** A hospital requires there to be 2 nurses for every 8 patients. If the hospital has a total of 80 patients and nurses on one of the floors, how many nurses and how many patients are there? Show or explain your thinking.
- 8.** A teacher is planning a class trip to the zoo. The zoo requires 3 adults to join for every 10 students. If the teacher orders 78 tickets, how many tickets are for adults and how many are for students? Show or explain your thinking.

Science Mom Lesson 22

Name: Date: Period:

Additional Practice

ACC6.2.13

Problems 1–3: For each pair, circle the larger unit of measurement.

- 1.** Mile Meter
- 2.** Inch Foot
- 3.** Pound Gram

4. For each unit of measurement, select whether it measures length, volume, or weight.

Unit	Length	Volume	Weight
Millimeter			
Pound			
Cup			
Quart			
Yard			
Gram			

Problems 5–7: Identify a unit that can be used to measure:

5. The length of a hammer.
6. The volume of an orange soda can.
7. The weight of a truck.

8. For each pair, select the unit you would most likely use to measure a box of cereal.

- | | |
|------------------|-----------|
| a Inches | Miles |
| b Gallons | Cups |
| c Ounces | Kilograms |

9. Match each object with the unit you would most likely use to measure it.

- | | |
|--|-------------------|
| a. The height of a house. | Kilometers |
| b. The length of a nail. | Pounds |
| c. The weight of a staple. | Grams |
| d. The distance between two towns. | Centimeters |
| e. The weight of a piggy bank. | Feet |
| f. The volume of a pitcher of lemonade. | Gallons |

Additional Practice

ACC6.2.14

1. Use the conversion rate 1 inch $\approx$ 2.54 centimeters to complete the table.

Inches	Centimeters
1	2.54
	15.24
	254
	127
20	

2. There are 60 seconds in 1 minute. Use this conversion rate to complete the table.

Seconds	Minutes
60	100
	0.5
	3.5
	5
480	

3. There are 3,785 milliliters in 1 gallon, and there are 4 quarts in 1 gallon. Determine the following. Show or explain your thinking.

- a How many quarts are in 20 gallons?
- b How many milliliters are in 20 quarts?

4. There are 5,280 feet in 1 mile. A dolphin swam 3 miles. How many feet did the dolphin swim?

5. Clare is making spaghetti sauce that calls for 46 oz of tomatoes. There are 16 ounces in one pound. Determine the approximate weight of the tomatoes, in pounds.
6. A large jar holds 4 gallons of milk, which is about the same as 15.2 liters of milk. A smaller jar holds 2 gallons of milk. About how many liters does the small jar hold?
- A. 3.8 liters
 - B. 7.6 liters
 - C. 11.4 liters
 - D. 30.4 liters
7. Noah's recipe for bread calls for 8 pounds of flour. He has a scale that measures in kilograms. 1 kilogram is approximately 2.2 pounds.
- a About how many kilograms of flour does Noah need?
 - b Noah also has a 1-cup measuring cup. If it takes about 4 cups of flour to equal 1 pound, how many cups of flour does Noah need?
8. There are 16 tablespoons in 1 cup. Priya says that $1\frac{1}{4}$ cups of chopped onions is equivalent to 60 tablespoons. Is Priya correct? Explain your thinking.

Additional Practice

ACC6.2.16

Problems 1–2: Turtle A walks 1.5 feet in 5 seconds. Turtle B walks 3 feet in 9 seconds. Each turtle keeps walking at those speeds.

1. How far does each turtle walk in 45 seconds?

2. If the two turtles start at the same place and walk in the same direction, how far apart will the two turtles be after 3 minutes? Show or explain your thinking.

3. A cheetah runs 100 meters in 6 seconds.
 - a At this rate, how long will it take the cheetah to run 150 meters?

 - b How far will the cheetah run in 27 seconds?

4. Snail A travels 4 inches in 7 minutes. Snail B travels 6 inches in 10 minutes. Each snail continues traveling at a constant speed.
 - a How far does Snail A travel in 35 minutes? Show your thinking.

 - b How far does Snail B travel in 35 minutes? Show your thinking.

 - c If the two snails start at the same place and travel in the same direction, how far apart will the two snails be after 84 minutes?

5. Elena reads 15 pages in 25 minutes. Shawn reads 12 pages in 15 minutes. Both read at a constant rate.

- a How many pages can Elena read in 1 hour?
- b How many pages can Shawn read in 1 hour?
- c Who reads more quickly?

Problems 6–8: Priya types 1,100 words in 20 minutes. Tyler types 600 words in 10 minutes. Andre types 300 words in 6 minutes. Each person types at a constant rate.

6. Complete the tables to represent the number of words each person can type in different amounts of time.

Priya		Tyler		Andre	
Words	Time (min)	Words	Time (min)	Words	Time (min)
	1		1		1
	30		30		30
	45		45		45

7. Which person is typing at the fastest rate? Show or explain your thinking.
8. How long would it take Priya to type 3,300 words? Explain your thinking.

Additional Practice

ACC6.2.17

Problems 1–3: Tomas and Leslie purchased a new printer for their office. The printer can print 250 pages every 2 minutes.

1. How many pages can the printer print per minute?

2. How many minutes does it take to print one page?

3. How many pages can be printed in 10 minutes?

4. A farm lets you pick 2 pints of blueberries for \$6.00.

- a What is the cost per pint?
- b How many pints can you buy per dollar?
- c At this rate, how many pints can you buy for \$15?
- d At this rate, how much will 7 pints of blueberries cost?

5. A farm lets you pick 4 pounds of strawberries for \$16.00.

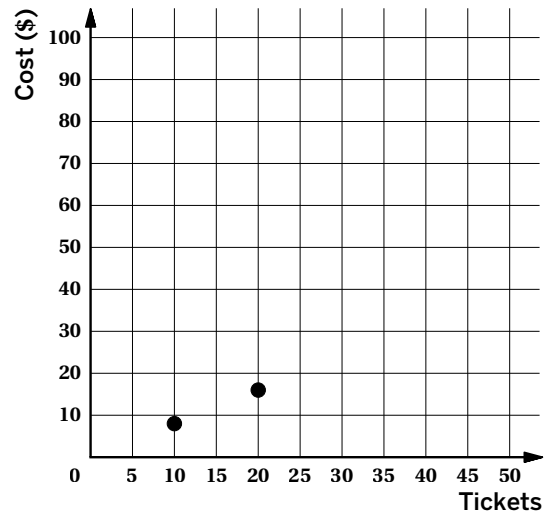
- a What is the cost per pound of strawberries?
- b How many pounds of strawberries can you buy per dollar?
- c At this rate, how many pounds can you buy for \$22?
- d At this rate, how much will 10 pounds of strawberries cost?

Problems 6–8: Clare paid \$10 for 8 game tickets at her school carnival.

- 6.** Complete the table to show the cost of different numbers of tickets.

Tickets	Cost (\$)
8	10
1	
10	
	18.75
	30

- 7.** Clare says that this graph represents two pairs of numbers of tickets and cost in dollars. Is Clare's graph correct? Explain your thinking.



- 8.** Kiran says that if he pays \$6 for 4 tickets, he's getting a better deal than Clare. Is Kiran correct? Explain your thinking.

Additional Practice

ACC6.2.18

1. A train travels 8 miles in 6 minutes. Which of these is *not* an equivalent rate?
 - A. 4 miles in 3 minutes
 - B. 1 mile in 45 seconds
 - C. $\frac{4}{3}$ miles in 1 minute
 - D. $\frac{3}{4}$ miles in 1 minute

2. A 2-pound squash costs \$8 at a grocery store. Which of these represent the same price per pound of squash? Select *all* that apply.
 - ☐ A. A 1-pound squash for \$4
 - ☐ B. A 1-pound squash for \$2
 - ☐ C. A $\frac{1}{4}$ -pound squash for \$1
 - ☐ D. A 1-pound squash for \$0.25
 - ☐ E. 8 pounds of squash for \$32

3. Lin bought a 40-pound bag of dog food for \$48.80. Shawn bought an 8-pound bag of dog food for \$9.88.

- a Complete this table to determine the unit rate for each bag of dog food.

	Dog food (lb)	Price (\$)	Unit rate (\$ per lb)
Lin	40	48.80	
Shawn	8	9.88	

- b Which size bag of dog food offers the best deal? Explain your thinking.

4. Which brand of peanut butter offers the best deal? Explain your thinking.
 - Brand A: 28 ounces for \$4.76
 - Brand B: 48 ounces for \$8.64
 - Brand C: 16 ounces for \$3.52

5. A grocery store sells four kinds of apples in bags.

- a** Complete the table to determine the unit price for each variety of apple.

Apple Variety	Price (\$)	Weight (lb)	Unit Rate (\$ per lb)
Gala	6.45	5	
Fuji	3.08	4	
Honeycrisp	5.37	3	
Granny Smith	7.74	6	

- b** Which type of apple is the best deal? Explain your thinking.

6. Jada reads 15 pages in 12 minutes. Bard reads 21 pages in 28 minutes. Who reads faster? Explain your thinking.

7. Mai uses 8 cups of apples to make 3 pies. Mai says, “At this same rate, I can make 5 pies using 16 cups of apples.” Is Mai correct? Explain your thinking.

8. Plane A travels 2,260 miles in 4 hours. Plane B travels 810 miles in $1\frac{1}{2}$ hours. Plane C travels 3,692 miles in $6\frac{1}{2}$ hours. Tyler claims that Plane C has the fastest speed. Is Tyler correct? Explain your thinking.

Additional Practice

ACC6.3.01

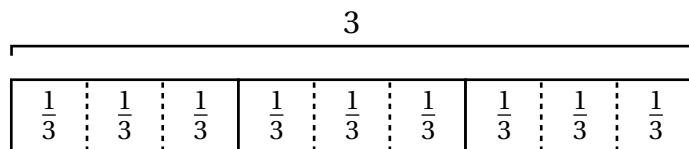
1. A deli makes 10 pounds of potato salad. If the deli sells portions of potato salad in $\frac{1}{2}$ -pound containers, how many portions can the deli sell?
 - a Draw a tape diagram to model this scenario.

- b Write a division expression that represents this scenario.

- c How many portions of potato salad can the deli sell?

2. Refer to the tape diagram shown.

- a Write a multiplication equation that could be represented by the diagram.



- b Write a division equation that could be represented by the diagram.

3. Andre's neighborhood swimming pool is open for 10 hours each day. Each lifeguard's shift is $2\frac{1}{2}$ hours. How many shifts will be available throughout the day?

4. Mai was filling several same-sized jars with oatmeal to make overnight oats. She put $\frac{2}{3}$ cups of oatmeal in each jar, using a total of 6 cups.

- Mai says, "I can use the equation $6 \cdot ? = \frac{2}{3}$ to determine how many jars are filled."
- Diego says, "I can use the equation $6 \div \frac{2}{3} = ?$ to determine how many jars are filled."

Who is correct? Circle one.

Mai Diego Both Neither

Explain your thinking.

Problems 5–6: Philippe and his two friends Angelica and Tomas are each making themselves an ice cream sundae using Philippe's recipe. Angelica and Tomas are using different-sized scoops to make their sundaes. If Philippe's recipe calls for 6 cups of ice cream, how many scoops of ice cream does each friend need based on the size of their scoop?

5. Tomas: $\frac{1}{2}$ -cup scoop

6. Angelica: 3-cup scoop

7. Priya was filling snack bags with trail mix. After using 9 cups of trail mix, she had filled 12 bags. All of the snack bags have the same amount of trail mix.

a Write a multiplication equation to represent this situation.

b Write a division equation to represent this situation.

c Draw a diagram to represent this situation.

d How much trail mix is in each bag?

Additional Practice

ACC6.3.02

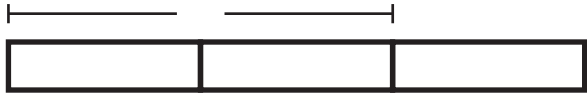
Problems 1–5: Levi is planting herbs in his garden. Determine how many of each herb plant Levi can fit in 1 planter. Use the diagrams if they help with your thinking.



1. Sage, if 10 sage plants fill $\frac{1}{2}$ of a planter.
2. Thyme, if 8 thyme plants fill $\frac{2}{3}$ of a planter.
3. Basil, if 3 basil plants fill $\frac{1}{3}$ of a planter.
4. Oregano, if 3 oregano plants fill $\frac{3}{4}$ of a planter.
5. Levi wrote the expression $3 \div \frac{3}{4}$ to represent how many oregano plants fill 1 planter. Describe a situation that represents the expression $6 \div \frac{2}{3}$.

Problems 6–7: Juni picks 12 blackberries from her backyard. The blackberries fill $\frac{2}{3}$ of a cup.

6. Label the tape diagram to represent Juni’s situation.



- 7.** Determine how many blackberries fill 1 cup. Use the tape diagram in problem 6 if it helps with your thinking.
- 8.** Maj is painting his kitchen. He uses 5 gallons of paint for $\frac{5}{8}$ of the kitchen. How many gallons of paint would he need to paint the whole kitchen?
- 9.** Sergio is planting flowers in his garden. 21 flowers fill 3 small planters. How many flowers fill 1 small planter?
- 10.** Match each representation with a question.

Representation	18 flowers fill 2 planters. How many flowers fill 1 planter?	18 flowers fill $\frac{2}{3}$ of a planter. How many flowers fill 1 planter?
$18 \div 2 = ?$		
$18 \div \frac{2}{3} = ?$		
$2 \times ? = 18$		
$\frac{2}{3} \times ? = 18$		

Additional Practice

ACC6.3.03

Problems 1–2: Think about the expression $5\frac{1}{4} \div \frac{3}{4}$.

1. Draw a tape diagram to represent the expression.

2. Calculate the quotient.

Problems 3–4: Think about the expression $2\frac{1}{5} \div \frac{2}{5}$.

3. Draw a tape diagram to represent the expression.

4. Calculate the quotient.

5. Bard buys rice in a 4-cup box. Bard's family uses $1\frac{1}{3}$ cups of rice per meal. How many meals does one box last?
 - a Draw a diagram to represent the scenario. Label your diagram.
 - b Write a multiplication or division equation to represent the scenario. Use a question mark for the unknown.
 - c Determine how many meals one box lasts. Explain your thinking.
6. Clare is cutting a 4-foot board into $\frac{4}{5}$ -foot sections. How many sections of board will Clare cut?
 - a Draw a diagram to represent the scenario. Label your diagram.
 - b Write a multiplication or division equation to represent the scenario. Use a question mark for the unknown.
 - c Determine how many sections of board Clare will cut.
7. Diego and Han were each asked to write a question that represents the equation? $\cdot \frac{1}{5} = 6$.

Diego's Response	Han's Response
How many $\frac{1}{5}$ s are in 6?	How many 6s are in $\frac{1}{5}$?

Who wrote a question that correctly represents the equation? Circle one.

Diego Han Neither

Explain your thinking.

Additional Practice

ACC6.3.04

1. Select *all* the expressions whose value is greater than 1.

☐ A. $2 \div \frac{3}{2}$

☐ B. $\frac{3}{2} \div 2$

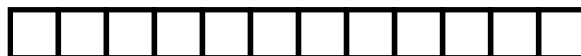
☐ C. $\frac{3}{4} \div \frac{1}{3}$

☐ D. $\frac{4}{3} \div 2$

☐ E. $4 \div \frac{3}{4}$

2. Jada ordered a 3-foot sub from the grocery store for a party. She cuts the sub into $\frac{1}{2}$ -foot servings. Jada says the sub is long enough to feed 6 people. Is Jada correct? Explain your thinking.

Problems 3–4: Here is a diagram.



3. Determine if the value of $1\frac{1}{2} \div \frac{2}{3}$ is:

Less than 1

Greater than 1

Exactly 1

4. Calculate the value of the expression in Problem 3.

Use the diagram if it helps you with your thinking.

Problems 5–6: Here is a diagram.



5. Determine if the value of $\frac{1}{2} \div \frac{3}{4}$ is:

Less than 1

Greater than 1

Exactly 1

6. Calculate the value of the expression in Problem 5.

Use the diagram if it helps you with your thinking.

Problems 7–9: Here is a diagram.



7. Calculate $\frac{1}{3} \div \frac{3}{2}$.

Use the diagram if it helps you with your thinking.

8. Calculate $\frac{3}{2} \div \frac{1}{3}$.

Use the diagram if it helps you with your thinking.

9. How are Problems 7 and 8 alike? How do their solutions compare? Why do you think this is the case?

10. Kiran describes how to draw a diagram to represent and calculate $\frac{4}{3} \div \frac{1}{2}$.

Kiran's Response

Draw a tape diagram whose length represents $\frac{4}{3}$. Partition the diagram into 4 equal parts to show 4 groups of $\frac{1}{3}$. Then, partition each $\frac{1}{3}$ into 2 equal parts.

There are 8 groups of $\frac{1}{2}$'s in $\frac{4}{3}$. Therefore, the value of $\frac{4}{3} \div \frac{1}{2}$ is 8.

Did Kiran correctly describe how to draw a tape diagram? Explain or show your thinking.

Additional Practice

ACC6.3.05

Problems 1–4: Calculate the value of each expression. Draw a diagram if it helps with your thinking.

1. $4 \div \frac{3}{4}$

2. $3\frac{2}{3} \div \frac{5}{6}$

3. $\frac{5}{2} \div 1\frac{1}{3}$

4. $2\frac{3}{4} \div \frac{5}{6}$

Problems 5–6: Noah picked $3\frac{1}{2}$ cups of blackberries, which is enough for $\frac{3}{4}$ jars of blackberry jam.

5. How many cups does Noah need for 1 jar of blackberry jam? Show or explain your thinking.

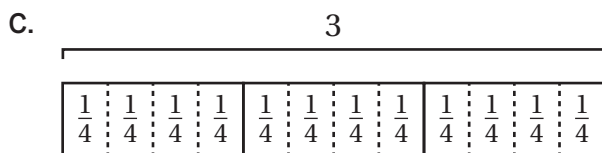
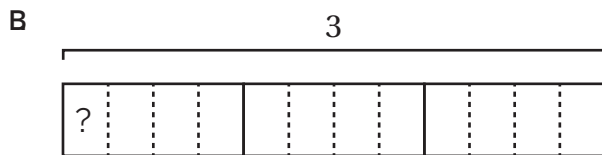
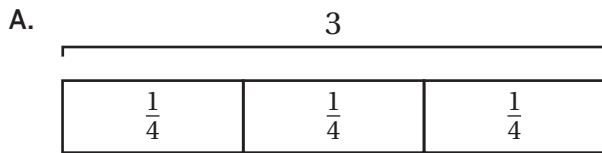
6. How many jars can Noah fill using 8 cups of blackberries? Show or explain your thinking.

- 7.** One batch of tomato sauce uses $3\frac{1}{2}$ pounds of tomatoes. André has 5 pounds of tomatoes. He says he can make $1\frac{3}{5}$ batches of tomato sauce. Is André correct? Explain your thinking.
- 8.** After charging for $1\frac{1}{3}$ hours, a laptop is at $\frac{3}{5}$ of its full power. How long will it take the laptop to charge completely? Show or explain your thinking.
- 9.** Shawn has $5\frac{1}{2}$ cups of lemonade. If each serving of lemonade is $\frac{3}{4}$ cups, how many servings does Shawn have?

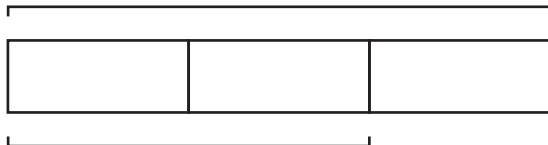
Additional Practice

ACC6.3.06

1. Which of the tape diagrams shown best represents the expression $3 \div \frac{1}{4}$?



2. Complete the tape diagram shown to represent the problem $1 \div \frac{2}{3}$. Then, calculate $1 \div \frac{2}{3}$.

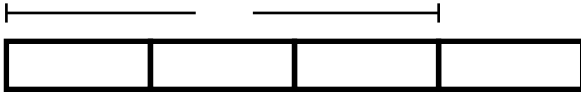


3. Tyler picked 1 cup of blueberries, which is enough for $1\frac{1}{2}$ batches of blueberry muffins. How many cups does he need for 1 batch? Draw a diagram if it helps with your thinking.

4. Shawn painted $\frac{5}{2}$ square yards of wall area with 2 gallons of paint. How many gallons of paint did Shawn need to paint each square yard of wall?

Problems 5–6: Complete the tape diagram to represent and solve each problem.

5. Trevor picked 1.5 cups of raspberries, which is enough for $\frac{3}{4}$ of a pan of raspberry shortcake. How many cups does he need for a whole pan of raspberry shortcake?

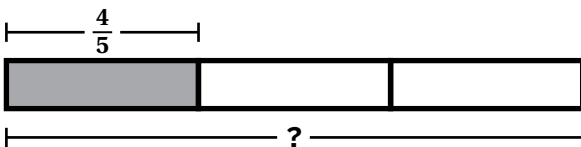


6. Tonya picked 6 cups of blueberries, which is enough for $\frac{3}{4}$ of a pan of blueberry crumble. How many cups does she need for a whole blueberry crumble?



7. $\frac{3}{5}$ of the town population walks to the park on a given Saturday. If 180 town residents walked to the park that day, how many total residents live in the town?

8. Calculate $\frac{4}{5} \div \frac{1}{3}$. Show or explain your thinking. Use the tape diagram if it helps with your thinking.



Additional Practice

ACC6.3.07

Problems 1–5: Use any strategy to calculate each quotient.

1. $9 \div \frac{1}{3}$

2. $9 \div \frac{3}{5}$

3. $4\frac{3}{4} \div \frac{3}{8}$

4. $\frac{1}{3} \div \frac{4}{5}$

5. $2\frac{3}{4} \div \frac{1}{2}$

6. How many groups of $\frac{3}{4}$ are in $6\frac{1}{2}$?

7. How many groups of $\frac{3}{4}$ are in $3\frac{2}{3}$?

8. Here are four expressions:

$$\frac{5}{3} \div \frac{2}{5}$$

$$\frac{5}{3} \div 2$$

$$\frac{5}{3} \div \frac{1}{5}$$

$$\frac{5}{3} \div 1$$

a Order these expressions by value from *least* to *greatest*.

--	--	--	--

Least

Greatest

b How are these expressions alike? How are these expressions different?

9. Use the equation $3\frac{1}{2} \div \frac{1}{6} = 21$ to determine $3\frac{1}{2} \div \frac{2}{6}$. Explain your thinking.

10. Calculate the value of the expression. Show or explain your thinking.

$$\frac{5}{3} \div \frac{1}{5}$$

11. Here is a list of expressions.

$6 \times \frac{1}{2}$	$\frac{1}{3} \div \frac{1}{9}$
$\frac{2}{3} \div \frac{1}{9}$	$\frac{4}{5} \div \frac{1}{5}$
$\frac{2}{3} \div \frac{1}{6}$	$\frac{1}{3} \div \frac{1}{5}$
$9 \times \frac{1}{3}$	$4 \times \frac{3}{2}$

- a Group the expressions into the appropriate column of the table. One of the expressions will not be used.

Value of 3	Value of 4	Value of 6

- b Explain why one of the expressions does not fit into any of the columns in the table.

Additional Practice

4.12

1. How many groups of $\frac{5}{6}$ are in each of the following quantities? Show your thinking.

a $\frac{2}{3}$

b $1\frac{1}{2}$

c $4\frac{1}{6}$

2. How many groups of $1\frac{3}{4}$ are in each of the following quantities? Show your thinking.

a $2\frac{1}{2}$

b 4

c $\frac{3}{4}$

3. Shawn's golf club is $3\frac{3}{4}$ ft tall. Shawn's hockey stick is $5\frac{1}{2}$ ft tall.

a What fraction of the golf club's height is the hockey stick? Show your thinking.

b How many times as tall is the hockey stick than the golf club? Show your thinking.

4. Jada is $5\frac{1}{4}$ ft tall. Her little sister is 4 ft tall. How many times taller is Jada than her sister?

A. $\frac{4}{5}$

B. $\frac{16}{21}$

C. $1\frac{1}{4}$

D. $1\frac{5}{16}$

Clare and Diego each went for a bike ride. Clare rode $1\frac{1}{2}$ mi, and Diego rode $2\frac{3}{4}$ mi. Use this information for Problems 5–6.

5. How many times as far as Clare did Diego ride?

a Write a division expression to represent this situation.

b Determine the solution. Show your thinking.

6. What fraction of Diego's distance did Clare ride?

a Write a division expression to represent this situation.

b Determine the solution. Show your thinking.

7. Andre uses $1\frac{5}{6}$ lb of tomatoes for a batch of homemade spaghetti sauce. He has $9\frac{1}{2}$ lb of tomatoes from his garden. Does Andre have enough tomatoes to make 6 batches of spaghetti sauce? Explain your thinking.

8. Mai has worked $3\frac{3}{4}$ hours of her $7\frac{3}{4}$ hour shift. She says that she has worked more than half of her shift. Is Mai correct? Explain your thinking.

Additional Practice

ACC6.3.10

1. A rectangular coffee table has a length of $46\frac{1}{2}$ inches, a width of w inches, and an area of 1,395 square inches. Select *all* the equations that represent the relationship between the dimensions of the coffee table.

☐ A. $w \cdot 1395 = 46\frac{1}{2}$

☐ B. $w \div 1395 = 46\frac{1}{2}$

☐ C. $46\frac{1}{2} \cdot w = 1395$

☐ D. $46\frac{1}{2} \cdot 1395 = w$

☐ E. $1395 \div w = 46\frac{1}{2}$

☐ F. $1395 \div 46\frac{1}{2} = w$

2. The triangle has a height of 8 feet and an area of 42 square feet. Select *all* the equations that represent the relationship between the dimensions of the triangle.

☐ A. $42 = \frac{1}{2} \cdot b \cdot 8$

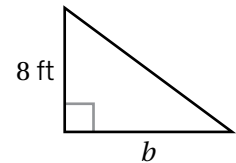
☐ B. $42 \div b = \frac{1}{2} \cdot 8$

☐ C. $42 \div b = 8$

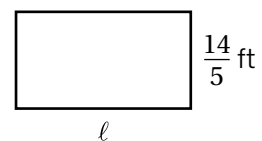
☐ D. $4 \cdot b = 42$

☐ E. $8 \cdot b = 42$

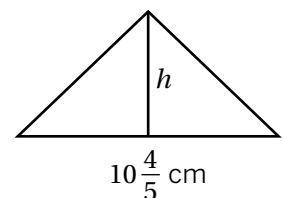
☐ F. $b \div 8 = 42$



3. The area of the rectangle is 28 square feet. What is the length of the rectangle? Show your thinking.

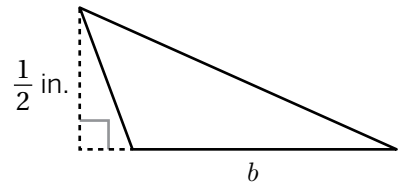


4. The area of the triangle is 27 square centimeters. What is the missing height h ? Show your thinking.

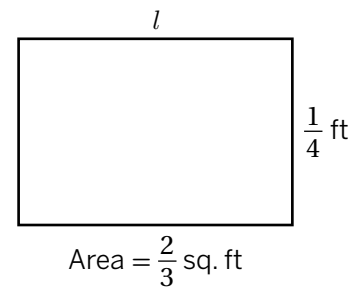


5. Han is tiling a shower wall that is 8 feet by 6 feet. The tiles are squares with a side length of $1\frac{1}{3}$ feet. How many tiles are needed to cover the entire shower wall? Show or explain your thinking.

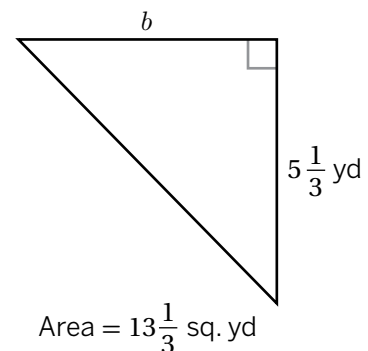
6. The area of the triangle is $\frac{2}{7}$ square inches.
What is the missing length b ? Show your thinking.



7. Lin wrote the equation $\frac{2}{3} = l \cdot \frac{1}{4}$ to determine the missing length of the rectangle. Use Lin's equation to solve for l .



8. Bard wrote the equation $\frac{1}{2} \cdot b \cdot 13\frac{1}{3} = 5\frac{1}{3}$ to determine the unknown base of the triangle. Is Bard's equation correct? If yes, solve the equation to determine the unknown base. If no, write and solve a new equation for b .



Additional Practice

ACC6.3.12

1. Select *all* the expressions that have a value greater than 1.

☐ A. $0.61 + 0.25$

☐ B. $0.8 + 0.03$

☐ C. $0.85 + 0.36$

☐ D. $0.53 + 0.55$

☐ E. $0.1 + 0.098$

2. This diagram represents a square backyard with a garden and a pond. The shaded area represents the pond, and the unshaded area represents the garden. Each small square represents 1 hundredth of the total backyard area.

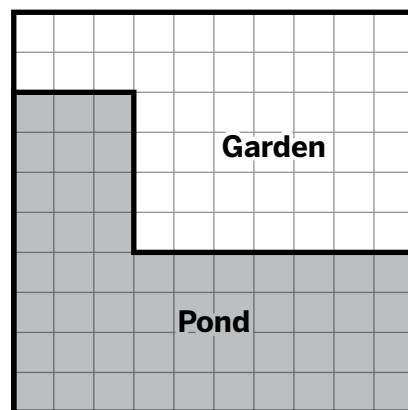
Which equation represents the difference between the size of the pond and the garden?

A. $1 - 0.52 = 0.48$

B. $0.48 + 0.52 = 1$

C. $0.52 - 0.48 = 0.04$

D. $1 + 0.48 = 1.48$



Problems 3–4: Determine each sum. Use a hundredths chart if it helps with your thinking.

3. $0.26 + 0.725$

4. $38.70 + 0.212$

5. Determine the sum: $2.6 + 0.31$. Show or explain your thinking.

6. Determine the difference: $5.972 - 2.87$. Show or explain your thinking.

7. Mai wrote a vertical calculation for the expression $8.75 + 3.1$. Her work is shown.

$$\begin{array}{r} 8.75 \\ + 3.01 \\ \hline 11.76 \end{array}$$

Did Mai correctly determine the sum? Explain your thinking.

8. Priya wrote a vertical calculation for the expression $0.91 - 0.8$. Her work is shown.

$$\begin{array}{r} 0.91 \\ - 0.8 \\ \hline 0.83 \end{array}$$

Did Priya correctly determine the difference? Explain your thinking.

Science Mom Lesson 37

Name: Date: Period:

Additional Practice

ACC6.3.13

- Determine which decimals have the same value as 0.81. Select *all* that apply.
☐ A. 0.80
☐ B. 0.801
☐ C. 0.810
☐ D. 0.8010
☐ E. 0.8100
- Determine which expressions have the same value as $1.9 + 0.653$. Select *all* that apply.
☐ A. $1.9 + 6.5$
☐ B. $1.9 + 0.653$
☐ C. $1.009 + 0.653$
☐ D. $1.900 + 0.653$
☐ E. $1.9000 + 0.653$
- Determine which expressions have the same value as $0.05 - 0.0036$. Select *all* that apply.
☐ A. $0.0500 - 0.0036$
☐ B. $0.0005 - 0.0036$
☐ C. $0.050 - 0.003$
☐ D. $0.05 - 0.36$
☐ E. $0.05000 - 0.00360$
- Determine each sum. Use a hundredths chart if it helps with your thinking.
a $0.513 + 0.168$ **b** $0.951 + 0.04$ **c** $6.4 + 0.399$
- Determine each difference. Use a hundredths chart if it helps with your thinking.
a $310.8 - 6.3$ **b** $14.39 - 4.1$ **c** $1.8 - 0.54$

Problems 6–7: The table shows the times, in seconds, of the top three teams in a snowboarding competition.

Rank	Medal	Run 1	Run 2
1	Gold	18.865	15.916
2	Silver	15.867	19.02
3	Bronze	16.56	18.81

6. Each team had two runs. Determine each team's total time by adding their times for the two runs.
- a Gold
 - b Silver
 - c Bronze
7. Determine the difference in times for each team's two runs by subtracting their fastest time from their slowest time.
- a Gold
 - b Silver
 - c Bronze
8. Diego wants to subtract 1.89 from 3, but he is not sure how to proceed because 3 is a whole number and 1.89 is a decimal. Determine the difference, and explain how Diego could solve this problem.

Additional Practice

ACC6.3.15

1. Select *all* expressions that are equivalent to $(0.5) \cdot (0.8)$.

☐ A. $5 \cdot (0.1) \cdot 8 \cdot (0.01)$

☐ B. $5 \cdot (0.01) \cdot 8 \cdot (0.1)$

☐ C. $5 \cdot (0.1) \cdot 8 \cdot (0.1)$

☐ D. $5 \cdot (0.01) \cdot 8 \cdot (0.01)$

☐ E. $5 \cdot \frac{1}{10} \cdot 8 \cdot \frac{1}{10}$

☐ F. $5 \cdot 8 \cdot \frac{1}{10} \cdot \frac{1}{10}$

☐ G. $5 \cdot 8 \cdot \frac{1}{10}$

☐ H. $\frac{5}{10} \cdot \frac{8}{10}$

2. Determine each product.

a $15 \cdot (0.1)$

b $38 \cdot \frac{1}{10}$

c $3.1 \cdot 0.1$

d $1.84 \cdot \frac{1}{10}$

3. Determine each product.

a $151 \cdot \frac{1}{100}$

b $8.3 \cdot (0.01)$

c $1.95 \cdot \frac{1}{100}$

d $9.436 \cdot (0.01)$

4. Which expressions have the same value as $(0.316) \cdot (0.9)$? Select *all* that apply.

☐ **A.** 0.02844

☐ **B.** $316 \cdot 9 \cdot (0.0001)$

☐ **C.** $316 \cdot \frac{1}{1000} \cdot 9 \cdot \frac{1}{10}$

☐ **D.** $316 \cdot \frac{1}{1000} \cdot 9 \cdot \frac{1}{100}$

☐ **E.** $316 \cdot 9 \cdot \frac{1}{10000}$

5. Calculate the value of each expression. Show your thinking.

a $(5.7) \cdot 3$

b $(0.8) \cdot (6.4)$

c $(0.1) \cdot (0.23)$

d $(0.48) \cdot (0.29)$

6. Calculate the value of each expression. Show your thinking.

a $(1.6) \cdot (0.006)$

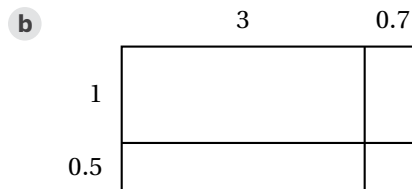
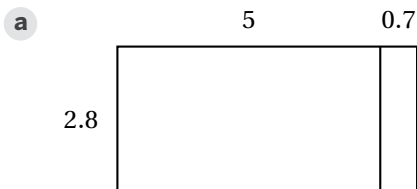
b $(0.007) \cdot (0.038)$

7. Noah calculated the value of the expression $(0.18) \cdot (0.09)$ by first writing the decimal factors as fractions, and then writing their product as a decimal. Noah's work is shown: $\frac{18}{100} \cdot \frac{9}{10} = \frac{162}{1000} = 0.162$. Is Noah correct? Explain your thinking.

Additional Practice

ACC6.3.16

1. Write a multiplication expression that represents each area diagram.



2. The rectangle has an area in square units and has been partitioned into four smaller rectangles. For each expression, write the name of the smaller rectangle whose area matches that expression. Then determine the area of each smaller rectangle.

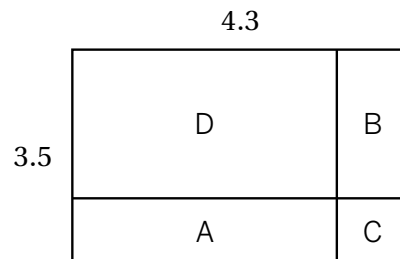
a $(0.3) \cdot 3$

b $3 \cdot 4$

c $(0.3) \cdot (0.5)$

d $4 \cdot (0.5)$

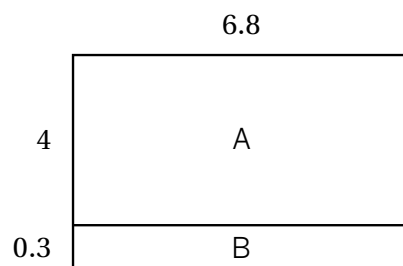
e Determine the total area of the rectangle.



3. The area diagram represents $(6.8) \cdot (4.3)$.

a Determine the areas of Rectangles A and B.

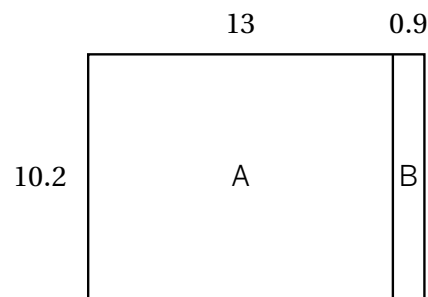
b What is the area of the larger 6.8 by 4.3 rectangle?



4. The area diagram represents $(13.9) \cdot (10.2)$.

a Determine the areas of Rectangles A and B.

b What is the area of the larger 13.9 by 10.2 rectangle?



5. Consider the expression $(0.24) \cdot (0.13)$.

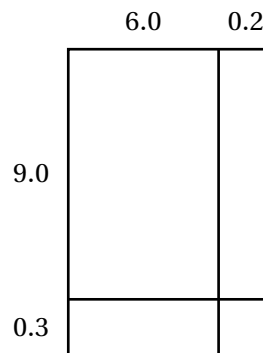
- a Draw an area diagram to represent this expression.
- b Determine the product of the expression. Use your area diagram if it helps with your thinking.

6. Consider the expression $(3.5) \cdot (1.1)$.

- a Draw an area diagram to represent this expression.
- b Determine the product of the expression. Use your area diagram if it helps with your thinking.

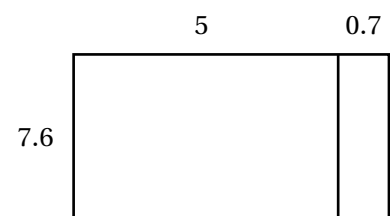
7. Tyler drew an area diagram to represent the expression $(6.2) \cdot (9.3)$.

- a Tyler's teacher says that there is another way he could use an area diagram to represent the same expression. Draw another area diagram that Tyler could use.



- b Determine the product of the expression. Use an area diagram if it helps with your thinking.

8. Bard drew this area diagram to represent the expression $(5.7) \cdot (7.6)$, and claimed to be able to use it to determine the area of the rectangle. Kiran said Bard's area diagram is not correct because it only shows two partitioned rectangles, when instead it should show four partitioned rectangles. Who is correct? Explain your thinking.



Additional Practice

ACC6.3.17

1. Determine the value of each expression.

a $3 \cdot (0.5)$

b $(6.2) \cdot 3$

c $2 \cdot (16.3)$

d $(0.9) \cdot 5$

2. A pound of turkey lunch meat at the deli costs \$7.49 per pound. Lin wants to purchase 2.25 pounds of turkey for the week. Determine how much Lin will pay, rounded to the nearest cent. Show your thinking.

Problems 3–6: Determine the value of each expression using any strategy. Show or explain your thinking.

3. $(3.4) \cdot (7.6)$

4. $(1.2) \cdot (9.9)$

5. $(0.54) \cdot (0.8)$

6. $(8.79) \cdot (6.04)$

- 7.** A dog weighs 5.4 kilograms and a cat weighs 10.25 pounds. Which weighs more and by how much? Note: 1 kilogram = 2.2 pounds. Show your thinking.
- 8.** A pound of strawberries costs \$4.29 and a pound of bananas costs \$0.69. What is the total cost of 1.5 pounds of strawberries and 0.8 pounds of bananas? Round your answer to the nearest cent. Show your thinking.
- 9.** Three polygons with their dimensions are given.
- A rectangle with a base of 6.437 centimeters and a height of 5.5 centimeters.
 - A square with side lengths of 5.99 centimeters.
 - A rectangle that is 14.25 centimeters wide and 2.5 centimeters long.
- Noah says that the square has the greatest area. Is Noah correct? Explain your thinking.

- 10.** There are 2.75 grams of sugar in 1 ounce of applesauce. Shawn says this means that there are 165 grams of sugar in 6 ounces of applesauce. Is Shawn correct? Explain your thinking.

Additional Practice

ACC6.3.18

1. Match each division problem with its quotient.

Division Expression	Decimal
a. $1 \div 4$	 0.8
b. $4 \div 5$	 0.3
c. $3 \div 10$	 0.5
d. $4 \div 8$	 0.25

2. Use long division to show that each equation is true.

a. $7 \div 10 = 0.7$ b. $9 \div 50 = 0.18$ c. $12 \div 25 = 0.48$

3. Use long division to determine each quotient. Write your answer as a decimal.

a. $26 \div 5$

b. $75 \div 8$

c. $79 \div 4$

4. Use long division to determine each quotient. Write your answer as a decimal.

a. $324 \div 5$

b. $509 \div 8$

c. $951 \div 6$

5. Use long division to determine each quotient. Write your answer as a decimal.

a $240 \div 32$

b $1650 \div 16$

c $8415 \div 20$

6. Salinity is a measurement of how much salt is in water. It is measured in grams of salt per liter of water. Use division to complete the table to show the salinity of each ocean.

Water source	Salt (g)	Water (L)	Salinity (g/L)
Indian Ocean	1,235	38	
Pacific Ocean	1,352	40	
Arctic Ocean	658	20	

7. Mai reasoned, " $\frac{22}{25}$ is equivalent to $\frac{44}{50}$ and to $\frac{88}{100}$, so the decimal of $\frac{22}{25}$ is 0.88."

a Use long division to show that Mai is correct.

b Use long division to determine whether the decimal of $\frac{44}{50}$ is also 0.88.

8. Noah calculated $1124 \div 20$. Priya said he made a mistake using long division. Is Priya correct? Explain your thinking.

Noah's work:

$$\begin{array}{r}
 5.62 \\
 20 \overline{)1124.00} \\
 \underline{-100} \\
 124 \\
 \underline{-120} \\
 40 \\
 \underline{-40} \\
 0
 \end{array}$$

Additional Practice

ACC6.3.19

1. Select *all* the expressions that are equivalent to $35 \div 10$.

- ☐ A. $0.35 \div 10$
- ☐ B. $3.5 \div 1$
- ☐ C. $350 \div 100$
- ☐ D. $350 \div 1,000$
- ☐ E. $3500 \div 100$

2. Select *all* the expressions that are equivalent to $2100 \div 30$.

- ☐ A. $210 \div 0.3$
- ☐ B. $21 \div 3$
- ☐ C. $2.1 \div 0.03$
- ☐ D. $21 \div 0.3$
- ☐ E. $210 \div 3$

3. Calculate $560.7 \div 5$. Show or explain your thinking.

4. Calculate $4295.9 \div 7$. Show or explain your thinking.

5. Use long division to determine each quotient.

a $15.4 \div 28$

b $6.86 \div 2$

c $529 \div 0.5$

6. Consider the expression $84 \div 1.5$.

a Write a related expression with the same value that uses only whole numbers.
Explain your thinking.

b Evaluate your expression.

7. Tyler paid \$37.36 for four 1-pound containers of protein powder. How much is each container of protein powder? Show your thinking.

8. Lin used long division to determine the quotient of $42 \div 1.4$. Her work is shown.
Is Lin *correct*? Explain your thinking.

$$\begin{array}{r} 0.3 \\ 1.4 \overline{)42} \\ \underline{-42} \\ 0 \end{array}$$

Additional Practice

ACC6.3.20

1. Select *all* the expressions that are equivalent to $54.4 \div 1.7$.

☐ A. $544 \div 17$

☐ B. $540 \div 17$

☐ C. $5440 \div 170$

☐ D. $544 \div 0.17$

☐ E. $5.44 \div 0.17$

2. Consider the expression $0.0063 \div 0.009$.

a Write two different division expressions that have the same quotient as $0.0063 \div 0.009$.

b Evaluate $0.0063 \div 0.009$. Show or explain your thinking.

3. Bard said, "To determine the value of $5.064 \div 8$, I can divide 5,064 by 80."

a Do you agree with Bard? Explain your thinking.

b Calculate the quotient of $5.064 \div 8$. Show your thinking.

4. Clare said, "To determine the value of $730.2 \div 0.6$, I can divide 7,302 by 6.

a Do you agree with Clare? Explain your thinking.

b Calculate the quotient of $730.2 \div 0.6$.

5. A bag of quarters weighs 8.5 kilograms. Each quarter weighs 5.67 grams. Which is the best estimate for the number of quarters in the bag? Show or explain your thinking.
(1 kilogram = 1000 grams)

A. 15,000

B. 1,500

C. 150

D. 15

6. Different brands of water bottles containing 16.9 ounces are sold in different quantities for different prices at a local grocery store. Complete the table to determine the unit cost for each brand.

Brand	Price (\$)	Quantity	Unit cost (\$)
Purely Water	4.40	40	
Spring Water	4.80	32	
Mountain Water	3.12	12	

- 7.** Evaluate each expression.

a $11.9 \div 1.7$

b $0.036 \div 0.02$

- 8.** Diego says, “To determine the value of $180 \div 1.2$, I can divide 18,000 by 120.” Mai says, “To determine the value of $180 \div 1.2$, I can divide 1,800 by 12.”

a Who is correct? Explain your thinking.

b Calculate the quotient of $180 \div 1.2$. Show or explain your thinking.

Additional Practice

5.16

- 1.** Circle the expression that has a greater value.

9% of 180

1.8% of 90

They have the same value.

Problems 2–5. Alicia sells art prints on her online shop. Every week, she sells \$550 of art prints. She tries to spend no more than 6% of her weekly art print sales on art supplies.

- 2.** Write an expression to represent how much money Alicia spends on art supplies every week.

- 3.** How much money, at most, does Alicia spend on art supplies each week?

4. Alicia puts 11% of her weekly art print sales into a savings account for an upcoming art fair. How much money does Alicia save each week?

5. Alicia recently had to spend \$99.00 to renew her website hosting platform. What percent of her weekly art print sales did she spend on website maintenance?

- A. 0.18%
- B. 9%
- C. 18%
- D. 90%

Problems 6–7. Capri is remodeling their kitchen. They went to the hardware store and purchased these items.

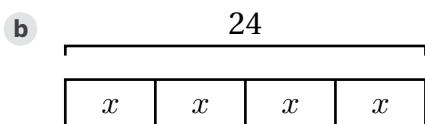
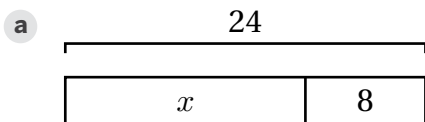
Items	Cost (\$)
Hammer	13.57
Wood	27.25
Nails	5.25
Screwdriver	9.50
Paint	12.66
Paintbrush	4.80
Floor tiles	35.97
Total	109.00

6. Floor tiles are the most expensive item. What percent of the total cost are the floor tiles?
- A. 30%
 - B. 31%
 - C. 32%
 - D. 33%
7. What percentage of the total hardware store bill is wood? Show or explain your thinking.

Additional Practice

ACC6.4.01

1. Determine the value of x in each tape diagram.



2. Which equations could represent the tape diagram?

Select *all* that apply.

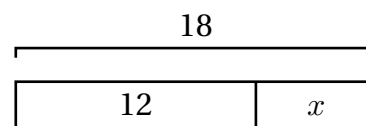
☐ A. $12 - x = 18$

☐ B. $12 + x = 18$

☐ C. $12 = 18 + x$

☐ D. $x = 18 + 12$

☐ E. $18 - x = 12$



3. Which equations could represent the tape diagram?

Select *all* that apply.

☐ A. $x + x + x + x + x = 10$

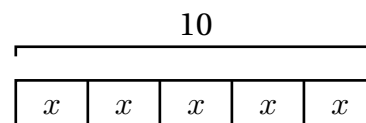
☐ B. $x = 10 \div 5$

☐ C. $5 \div 10 = x$

☐ D. $5 \cdot x = 10$

☐ E. $10 + 10 + 10 + 10 + 10 = x$

☐ F. $10 = x \cdot 5$

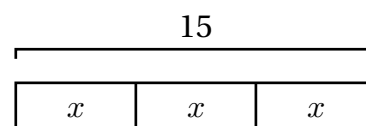


4. Consider the tape diagram shown.

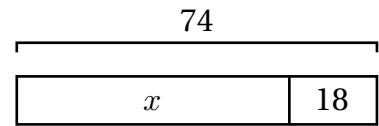
a Write an addition equation that represents the tape diagram.

b Write a multiplication equation that represents the tape diagram.

c Write a division equation that represents the tape diagram.



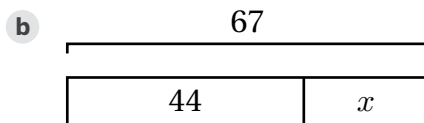
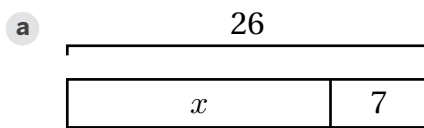
5. Consider the tape diagram shown



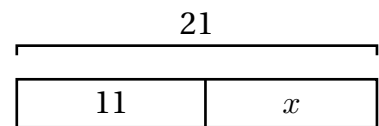
a Write an addition equation that represents the tape diagram.

b Write a subtraction equation that represents the tape diagram.

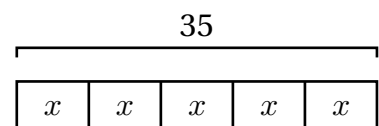
6. Determine the value of x in each tape diagram.



7. Consider the tape diagram shown. How does the diagram show that $x + 11$ has the same value as 21?



8. Consider the tape diagram shown. Diego says there are many equations that represents this tape diagram, such as $x + x + x + x + x = 35$. Is Diego correct? Explain your thinking.

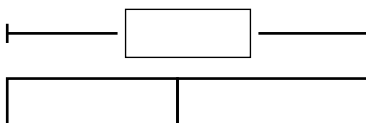


Additional Practice

ACC6.4.02

Problems 1–4: Willow and her friend made 34 ounces of hot chocolate. Willow drank 16 ounces of the hot chocolate. There are x ounces of hot chocolate left.

1. Complete the tape diagram to represent the situation.



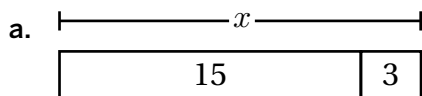
2. Select *all* the equations that could represent this situation.

- ☐ A. $34 + 16 = x$
- ☐ B. $34 - 16 = x$
- ☐ C. $16x = 34$
- ☐ D. $x + 16 = 34$
- ☐ E. $34 \div 16 = x$

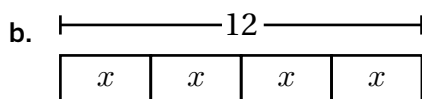
3. Determine the solution to one of the equations you selected in Problem 2.

4. Explain the solution's meaning in this situation.

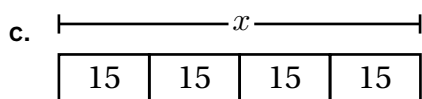
5. Match each equation to the tape diagram that represents it.



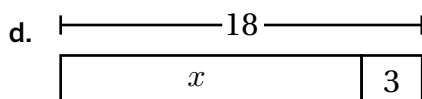
..... $18 - 3 = x$



..... $15 + 3 = x$



..... $x \div 4 = 15$



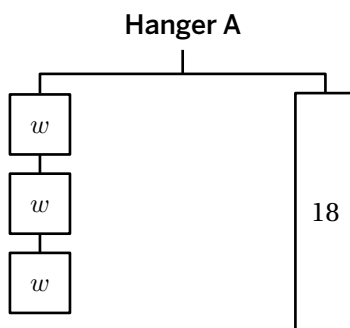
..... $4x = 12$

Additional Practice

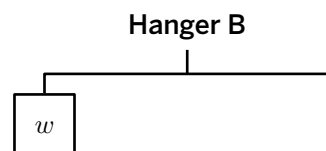
ACC6.4.03

Problems 1–2: Here are two hanger diagrams.

1. Hanger A is balanced. Write an equation that represents Hanger A.

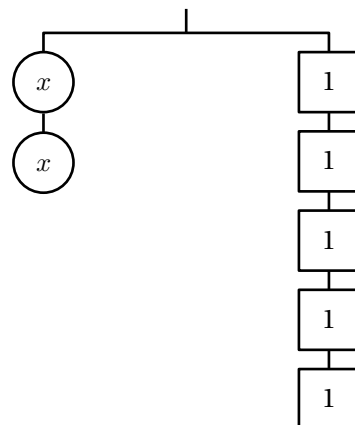


2. Determine the value of w that balances Hanger A. Use Hanger B if it helps with your thinking.



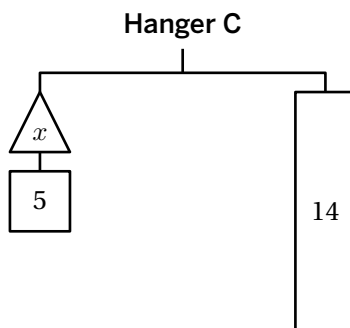
3. Which equations match the hanger diagram? Select *all* that apply.

- ☐ A. $2x = 1 + 1 + 1 + 1 + 1$
- ☐ B. $x + x = 1$
- ☐ C. $x + 3 = 5$
- ☐ D. $x + x = 5$
- ☐ E. $2x = 5$

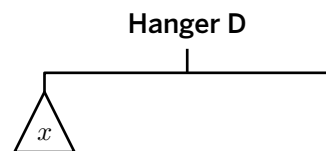


4. Here are Hangers C and D.

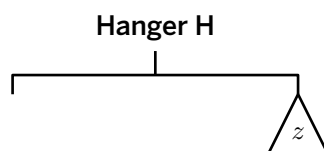
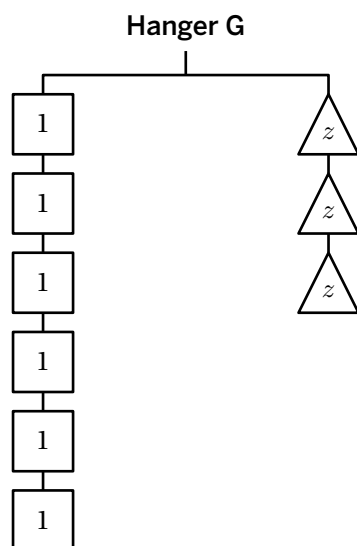
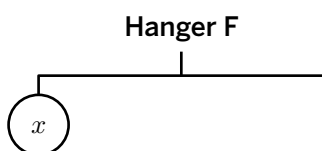
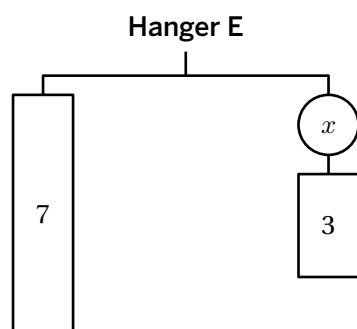
- a. Hanger C is balanced. Write an equation that represents Hanger C.



- b. Determine the value of x that balances Hanger C. Use Hanger D if it helps with your thinking.



Problems 5–7: Here are some balanced hangers.



5. Write equations that represent Hangers E and G.

Hanger E:

Hanger G:

6. Determine the values of x and z that balance Hanger E and Hanger G. Use Hanger F and Hanger H if they help with your thinking.

Hanger E:

Hanger G:

7. Shawn wants to create a balanced hanger by using only x and z . Write an equation that Shawn could use. Explain your thinking.

Additional Practice

ACC6.4.04

1. Write whether each equation is *true* or *false*.

a $11 + 17 = 14 + 14$

c $4\frac{1}{3} - 1\frac{2}{3} = 2\frac{1}{3}$

b $4 \cdot 6 = 25$

d $6 = 36 \div 6$

2. Match each equation with its solution.

Equation

Solution

a. $6.8 - a = 0.8$

..... $\frac{8}{3}$

b. $2b = 6.8$

..... 13.6

c. $\frac{3}{8} \cdot c = 1$

..... $1\frac{3}{8}$

d. $d \div 2 = 6.8$

..... 6

e. $e \div \frac{3}{8} = 1$

..... 4.8

f. $f + 2 = 6.8$

..... $\frac{3}{8}$

g. $g + \frac{5}{8} = 2$

..... 3.4

3. A bottle of ketchup had 44 ounces in it. After a family used k ounces, 18 ounces were left.

a Write an equation to represent this scenario.

b If you substitute 24 or 26 as the value for k , which value makes the equation true? Explain your thinking.

4. Clare split 144 beads among x friends. Each friend received 18 beads.

- a** Write an equation to represent this scenario.

- b** If you substitute 7 or 8 as the value for x , which value makes the equation true? Explain your thinking.

5. A group of five friends earn a total of \$50 raking leaves in their neighborhood. Each friend earns the same amount of money, x dollars. Which equation represents this scenario?

- A.** $5 + 50 = x$
- B.** $5 \cdot 50 = x$
- C.** $5 \cdot x = 50$
- D.** $x - 5 = 50$

6. A soccer team played 18 games in their season. The team won x games and lost 4 games. Select *all* the equations that represent this scenario.

- ☐ **A.** $18 = x - 4$
- ☐ **B.** $4 + x = 18$
- ☐ **C.** $x = 18 - 4$
- ☐ **D.** $18 - x = 4$
- ☐ **E.** $4 = 18 + x$

7. Is $x = 24$ a solution to the equation $\frac{3}{4}x = 18$? Explain your thinking.

8. Noah says that $x = 7.5$ is a solution to the equation $x + 6.1 = 13.4$. Is Noah correct? Explain your thinking.

Additional Practice**ACC6.4.05**

Problems 1–3: Izzy reads 2.5 books each month. After x months, Izzy has read 12.5 books.

1. Write an equation that could represent this situation.
2. Describe the meaning of the x in the situation.
3. How many months did it take for Izzy to read 12.5 books?

Problems 4–5: Here is an equation:

$$y + 3 = 27$$

4. Write a situation that the equation could represent.
5. Describe the meaning of the y in your situation.

- 6.** Arti buys 7 lemons to make a lemon custard pie. When he makes the pie, he finds that each lemon has 6 seeds. Select *all* the equations that represent the total number of lemon seeds, s .

☐ **A.** $s = 6 \div 7$

☐ **B.** $7s = 6$

☐ **C.** $7 + 60 = s$

☐ **D.** $s = 7 \cdot 6$

☐ **E.** $s \div 7 = 6$

Problems 7–9: Willow has \$85 to spend on paint at an art supply shop. It costs \$4.25 per tube of paint. Willow can buy p tubes of paint.

- 7.** Write an equation for the situation.

- 8.** Solve the equation for p .

- 9.** Describe the solution's meaning.

Additional Practice

ACC6.4.07

1. Diego has 21 sheets of paper in his notebook and s sheets have Diego's written class notes. In this scenario, what does the expression $21 - s$ represent?

Problems 2–3: The variable p represents the number of plants in a greenhouse.

2. A gardener adds 3 plants to the greenhouse. Write an expression to represent the new number of plants in the greenhouse.

3. What could the expression $\frac{1}{3}p$ represent in this situation?

4. Kiara wants to make homemade bread for her family and friends. She needs flour and water for a bread recipe. The directions state that the amount of flour should be 6 cups more than the amount of water. Complete the table to show how much water is needed for different amounts of flour.

Flour (cups)	Water (cups)
10	
14	
17	
x	

Problems 5–7: Strawberries cost \$4.50 per pound. How much would it cost to buy:

5. 3 pounds of strawberries?

6. 7 pounds of strawberries?

7. x pounds of strawberries?

Problems 8–10: Evaluate the expression $2d + 3$ for each value of d . Use the example provided in the table as a guide.

d	Value of $2d + 3$
6	$2(6) + 3 = 15$

8. $d = 7$

9. $d = 0.6$

10. $d = \frac{1}{2}$

Name: Date: Period:

Additional Practice

ACC6.4.08

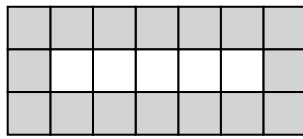
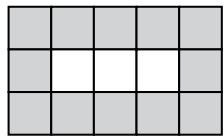
Problems 1–3: Here are six expressions.

$$3(x + 2) \qquad 3 + 6x \qquad x + x + x + 6$$

$$(3x + 2) + (3x + 2) \qquad x + 3 \qquad (2x + 3) + (x + 3)$$

1. Write all of the expressions that are equivalent to $3x + 6$.
2. Create another expression that is equivalent to $3x + 6$.
3. Choose an expression that is not equivalent to $3x + 6$. Explain how you know it is not equivalent.

Problems 4–6: Here are two unshaded t -by-1 rectangles with a tile border.



4. How many border tiles are there around the 3-by-1 rectangle?

5. How many border tiles are there around the 5-by-1 rectangle?

6. Tristan wants to create an expression to represent the number of border tiles needed to surround a t -by-1 rectangle. Which expression is correct?

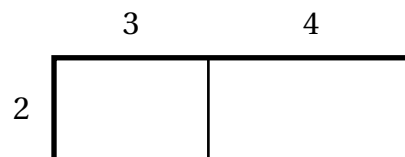
- A. $t + 6$
- B. $2t + 6$
- C. $t + 3$
- D. $2t + 3$

Additional Practice

ACC6.4.09

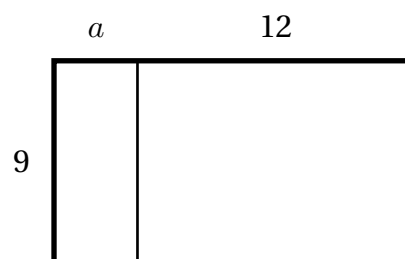
1. Select *all* the expressions that represent the area of the rectangle.

- ☐ A. $4 \cdot 2 + 4 \cdot 3$
☐ B. $2 \cdot 3 + 2 \cdot 4$
☐ C. $2 \cdot 3 + 4$
☐ D. $2(4 + 3)$
☐ E. $3(2 + 4)$



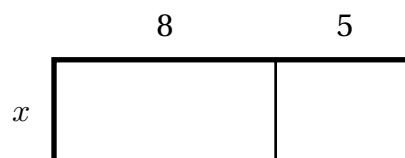
2. Select *all* the expressions that represent the area of the rectangle.

- ☐ A. $9a + 108$
☐ B. $12(a + 9)$
☐ C. $9 \cdot 12 + 12a$
☐ D. $9(a + 12)$
☐ E. $9a + 9 \cdot 12$

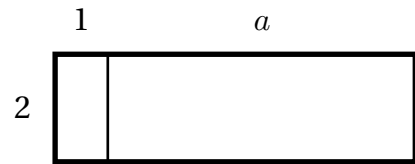


3. Select *all* the expressions that represent the area of the rectangle.

- ☐ A. $5x + 8x$
☐ B. $8x + 40$
☐ C. $x(5 + 8)$
☐ D. $5x + 40$
☐ E. $13x$



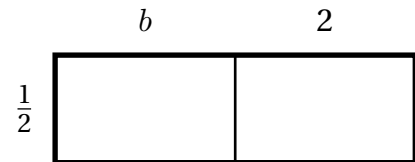
4. Write two equivalent expressions that could be used to represent the area of the rectangle.



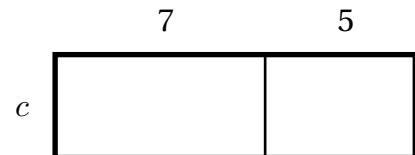
Expression 1:

Expression 2:

5. Bard writes the expression $\frac{1}{2}(b + 2)$ to represent the area of the rectangle. Write another expression that Bard could use to represent the same area.



6. Write two equivalent expressions that could be used to represent the area of the rectangle.



Expression 1:

Expression 2:

Additional Practice

ACC6.4.10

1. Rewrite each product expression as a sum expression.

a $3(x + 4)$

b $(5 + 7) \cdot x$

2. Rewrite each sum expression as a product expression.

a $6x + 9$

b $8x + 12y + 4z$

3. Select *all* the expressions that are equivalent to the expression $12x + 24$.

☐ **A.** $2(6x + 12)$

☐ **B.** $4(4x + 6)$

☐ **C.** $6(2x + 4)$

☐ **D.** $12(x + 2)$

☐ **E.** $24(2x + 1)$

4. Select *all* the expressions that are equivalent to the expression $14a - 28b + 42c$.

☐ **A.** $14(a - 2b + 3c)$

☐ **B.** $7(2a - 4b + 6c)$

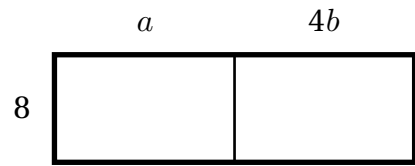
☐ **C.** $2(14a - 14b + 21c)$

☐ **D.** $14(a - 2b + 6c)$

☐ **E.** $7(2a - 4b + 7c)$

☐ **F.** $2(7a - 14b + 21c)$

5. Priya writes the area of this rectangle as the expression $8a + 32b$. Tyler writes the area of this rectangle as the expression $4(2a + 8b)$.



- a Who is correct? Explain your thinking.
- b Write two more equivalent expressions that represent the area of this rectangle.

6. Select *all* the expressions that are equivalent to the expression $3(8x + 6)$.

- ☐ A. $2(8x + 9)$ ☐ B. $6(4x + 3)$
- ☐ C. $6 + 24x$ ☐ D. $11x + 18$
- ☐ E. $24x + 18$

7. Clare rewrites the expression $12bc + 8bd$ as $b(12c + 8d)$.
Andre rewrites the expression $12bc + 8bd$ as $4(3bc + 2bd)$.

- a Are Clare's and Andre's expressions both equivalent to the expression $12bc + 8bd$?
Explain your thinking.
- b Write an additional equivalent expression to the expression $12bc + 8bd$.

Additional Practice

ACC6.4.11

1. Rewrite each expression using exponents.

a $9 \cdot 9 \cdot 9$

b $\frac{1}{2} \cdot \frac{1}{2}$

c $2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2$

2. Determine the value of each expression.

a 4^2

b 2^8

c $\left(\frac{1}{3}\right)^2$

3. Which value is equivalent to 8^2 ?

A. 6

B. 16

C. 32

D. 64

4. Using what you know about the properties of exponents, rewrite each expression using multiplication.

a 3^3

b 5^4

c 7^5

5. Select *all* the expressions that are equivalent to the expression 5^4 .

☐ A. 20^2

☐ B. $5 + 5 + 5 + 5$

☐ C. 4^5

☐ D. 625

☐ E. $5 \cdot 5^3$

☐ F. 25^2

6. The expression 3^5 is equal to 243. Use this to determine the value of each expression.

☐ a 3^6

☐ b $3^3 \cdot 3^2$

☐ c 3^4

7. The expression 6^4 is equal to 1,296. Use this to determine the value of each expression.

☐ a 6^3

☐ b $\frac{6^4}{6^2}$

☐ c $6^2 \cdot 6^2$

8. The number of blades of grass in a field is about 8.4 million. Is this number closer to 10^6 or 10^7 ? Explain your thinking.

Additional Practice

ACC6.4.12

1. Evaluate each expression.

a $12 + 3^2$

b $6^2 \cdot 5$

c $35 - 4^2$

d $100 - 7^2$

e $5^2 \div 5$

f $15 \cdot \left(\frac{1}{3}\right)^2$

2. Evaluate each expression.

a $10^3 - 8^2$

b $16 - 10^1$

c $20 + 2^4$

d $3^3 \cdot 5$

e $4 \cdot \left(\frac{1}{2}\right)^2$

f $\left(\frac{1}{6} \cdot 6\right)^6$

3. Determine whether the two expressions in each row are equivalent.
Write *equivalent* or *not equivalent* in the last column of the table.

Column A	Column B	Equivalent or not equivalent?
$8^2 + 10$	$10 + 2^6$	
$4^2 + 2^2$	$45 - 5^2$	
$(3 \cdot 4)^2$	$3^2 + 4^2$	
$9^3 + 9^2$	$3^4 \cdot 10$	
$25^2 + 400$	$10^3 - 25$	
$6 \cdot 4^1$	$12 \cdot 2^2$	

Additional Practice

ACC6.4.13

1. Determine the value of each expression when $x = 4$.

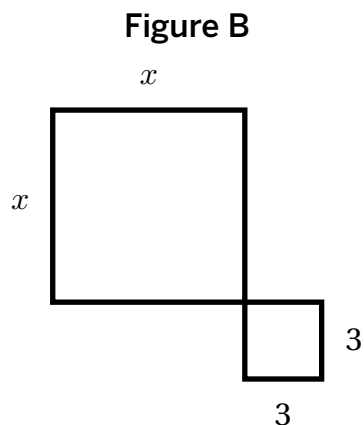
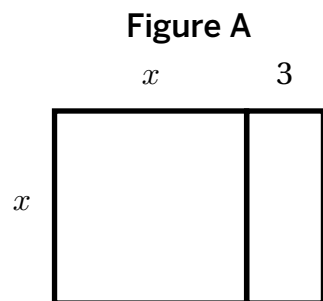
Expression	$2^x + 6$	x^3	$x^2 - 8$	$(x + 2)^2$
Value When $x = 4$				

2. Determine the value of each expression when $x = 3$.

Expression	$x^3 + 6x$	$5x^2$	$(x - 1)^5$	$7 + 2x^2$
Value When $x = 3$				

3. Determine the value of $3x^2 - 5x + 10$ when $x = 5$.

Problems 4–5: Here are two figures.



4. Match each figure with an expression that represents its area. One expression will not have a match.

$x^2 + 3x$

$(x + 3)^2$

$x^2 + 3^2$

5. Which of the three expressions in Problem 4 has the greatest value when $x = 5$?

6. Evaluate each expression for the given value of x .

a 0.3^x , when x is 3

b $x^2 + 6$, when x is 4

c $2x^2 + 3x$, when x is 5

d $8x + x^2$, when x is 4

7. Clare evaluated the expression $6x^2 - 12$ when x is 2. She says the value of the expression is 1,716. Is Clare correct? Explain your thinking.

Additional Practice

ACC6.4.14

Problems 1–3: A florist is making bouquets of flowers. He puts 12 flowers in each bouquet.

1. Complete the table.

Number of Bouquets, b	Number of Flowers, f
1	
2	
6	
	96

2. Which variable is the independent variable, b or f ? Explain your thinking.

3. Which of these equations represents the relationship? Circle one.

$b = 12f$ $f = 12b$ Neither

Explain your thinking

Problems 4–5: Adhira has a coupon for \$8 off any item at a store.

4. Complete the table. All values are in dollars.

Original Price, b	Price With Coupon, c
10	
20	
35	
	40

5. Which equation represents the relationship in the table?

- A. $p = 8c$
 B. $c = p - 8$
 C. $c = 8p$
 D. $p + 8 = c$

Problems 6–8: The table shows the number of robots a factory can create in different numbers of hours.

6. Write an equation to represent the relationship between the number of hours, h , and the number of robots, r , the factory produces.

Number of Hours, h	Number of Robots, r
1	6
2	12
6	36
15	90

7. Which variable is the dependent variable, r or h ? Explain your thinking.
8. Kris says that the factory could produce 100 robots in less than 24 hours. Is Kris correct? Explain your thinking.

Additional Practice

ACC6.4.15

Problems 1–3: Han, Elena, and Clare are training for a bike race, and they bike at a constant speed.

1. Complete the table.

Time (hours), t	Han's Distance (miles), h	Elena's Distance (miles), e	Clare's Distance (miles), c
0.25		2.5	
0.5	6		
2			30
3	36	30	45
4			

2. Write three equations that represent the distance d traveled, in miles, given the time t , in hours.

Han	Elena	Clare
_____	_____	_____

3. For your equations for Problem 2, which is the dependent variable and which is the independent variable?

Problems 4–7: Bard reads 35 pages in an hour.

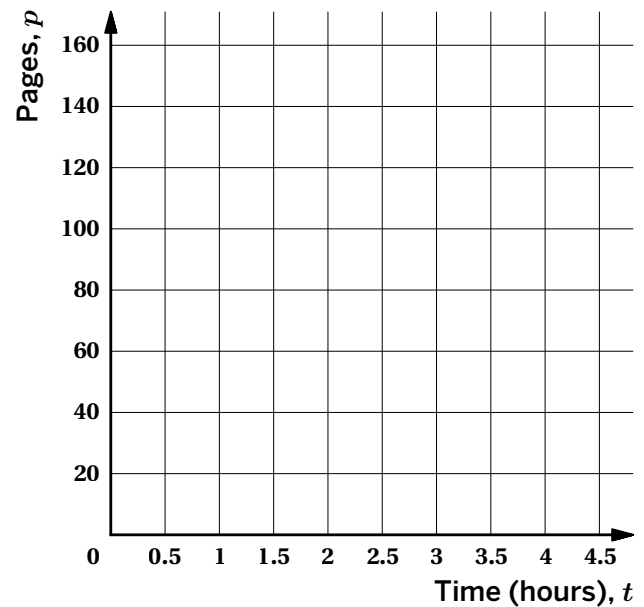
4. Assuming Bard reads at a constant rate, complete the table.

Time (hours), t	2	3.2	t		
Pages, p				140	56

5. Refer to the table.

- Write an equation that relates the number of pages read, p , to the time, t , in hours.
- Identify the independent and dependent variables in your equation.

- Use the points in the table to create a graph that shows the relationship between time, t , and the number of pages, p , Bard reads.
- Write an equation that Bard can use to determine the time, t , it will take to read 91 pages.



Additional Practice

6.15

1. Maria is making lemonade for a picnic with friends. The equation $w = l + 6$ represents the relationship between the amount of lemonade concentrate (in cups), l , and water (in cups), w . Which table represents the same relationship?

A.

l	w
8	2
10	4
12	6

B.

l	w
2	8
4	10
6	12

C.

l	w
10	4
8	2
6	0

D.

l	w
6	10
8	8
10	6

Problems 2–4: Sanjeev is buying limes for \$0.60 each.

2. Write an equation that represents how much money Sanjeev spends, m , for buying a number of limes, l .

3. Complete the table that represents this situation.

l	m
6	
7	
	\$5.40
	\$6.60

4. Explain what $l = 9$ and $m = \$5.40$ means in this situation.

5. Sanjeev spends \$9.00 on limes. How many limes did he purchase?

- A. 12 limes
- B. 13 limes
- C. 14 limes
- D. 15 limes

Additional Practice

ACC6.4.16

1. Select *all* the expressions that are equivalent to the expression $3a + 6a$.

☐ A. $4a + 5a$

☐ B. $a(3 + 6)$

☐ C. $6a + 3a$

☐ D. $a \div \frac{1}{18}$

☐ E. $9a$

☐ F. $18a$

☐ G. $3(a + 6a)$

☐ H. $a \cdot 9$

2. Determine the value of b . Show your thinking.

☐ a 45% of b is 225.

☐ b 300% of b is 12.

☐ c 12% of b is 4.8.

3. Solve each equation. Show your thinking.

☐ a $180 = 15c$

☐ b $3\frac{1}{5} + c = 7$

☐ c $c - 17.34 = 29.2$

☐ d $\frac{9}{8} = 2\frac{1}{2}c$

4. Select *all* the expressions that are equal to $5 \cdot 5 \cdot 5 \cdot 5$.

☐ A. $5 \cdot 4$

☐ B. $5^2 \cdot 5^2$

☐ C. $5^3 + 5$

☐ D. 4^5

☐ E. 5^4

☐ F. $5 \cdot 5^3$

5. Evaluate each expression. Show your thinking.

a $30 - 3^3$

b $5^3 + 16$

c $12^2 - 8^2$

d $8 \cdot \left(\frac{1}{2}\right)^3$

e $2^4 \cdot 2 = 32$

f $4^4 \div 4$

6. Select *all* the expressions that are equivalent to the expression $4(3x - 6)$.

☐ **A.** $3(4x - 6)$

☐ **B.** $2(6x - 12)$

☐ **C.** $8(2x - 3)$

☐ **D.** $6(2x - 4)$

☐ **E.** $12x - 24$

7. Shawn separately baked 6 batches of pretzels, for a total of 72 minutes. Each batch of pretzels baked for b minutes. Select *all* the equations that represent this situation.

☐ **A.** $6 \cdot b = 72$

☐ **B.** $b - 6 = 72$

☐ **C.** $b = 72 \div 6$

☐ **D.** $6 \cdot 72 = b$

☐ **E.** $72 = b + 6$

8. Priya says the solution to the equation $\frac{6}{5}x = \frac{1}{3}$ is $\frac{2}{5}$. Is Priya correct? Explain your thinking.

Additional Practice

ACC6.5.01

Problems 1–5: The table shows an original mixture for orange paint, as well as four other mixtures.

Mixture	Yellow Paint (cups)	Red Paint (cups)
Original	9	6
Mixture A	6	8
Mixture B	6	4
Mixture C	3	2
Mixture D	12	18

- Select *all* the mixtures that will create the same shade of orange as the original mixture.
 - ☐ A. Mixture A
 - ☐ B. Mixture B
 - ☐ C. Mixture C
 - ☐ D. Mixture D
- Choose one mixture that creates the same shade of orange as the original. Explain how you know it creates the same shade of orange as the original.
- Choose a mixture that does not have the same shade of orange as the original. How do the two mixtures compare?
- How much red paint would you need to mix with 1 cup of yellow paint to make a mixture that has the same shade of orange as the original mixture?
- Complete the table to create a different mixture that will also have the same shade of orange as the original mixture.

Yellow Paint (cups)	Red Paint (cups)

6. Brenden mixed 4 cups of blue paint with 3 cups of yellow paint to make the perfect color green to paint his bedroom.

Complete the table to show several other ways to make this color.

Yellow Paint (cups)	Blue Paint (cups)
3	4
9	
12	
	20

7. Select *all* the ratios that are equivalent to 3 : 4.

- ☐ A. 50 : 100
- ☐ B. 18 : 24
- ☐ C. $\frac{1}{4} : \frac{1}{5}$
- ☐ D. 75 : 100
- ☐ E. 4 : 5
- ☐ F. $\frac{1}{3} : \frac{1}{4}$

Additional Practice

ACC6.5.02

Problems 1–4: Complete the tables so that each relationship is proportional.

1.

x	y
16	4
20	
	9

2.

x	y
1	2.4
4	
	18

3.

x	y
12	36
1	
	0

4.

x	y
0.4	1
1	
	15

5. DesKayaks rents one-person and two-person kayaks for a fee of \$10 an hour plus \$2.50 per person for gear. The table shows the cost for several groups of people.

Is the relationship between the number of people renting kayaks and the fees charged a proportional relationship? Explain your thinking.

Number of People in Group	Total Fees (\$)
1	12.50
2	15.00
3	27.50
4	30.00

- 6.** Sabaan's favorite smoothie shop charges \$6.20 for 12 ounces. Complete the table so that it shows a proportional relationship between ounces of smoothie and cost. Show or explain your thinking.

Smoothie (oz)	Cost (\$)
12	6.60
18	
	13.20

- 7.** At an airport in Mexico City, Sara exchanged 300 U.S. dollars for 6,156 Mexican pesos. Complete the table so that it shows a proportional relationship between U.S. dollars and Mexican pesos. Show or explain your thinking.

U.S. Dollars	Mexican Pesos
300	6,156
1	
	10,260
	1

Science Mom Lesson 63

Name: Date: Period:

Additional Practice

ACC6.5.04

1. Diego is running at a speed of 3.5 meters per second. Complete the table.

Time (seconds)	Distance (meters)
1	3.5
15	
	70
60	
x	

2. The table shows a proportional relationship. Determine the constant of proportionality that completes the equation.

$$c = \dots\dots\dots w$$

w	c
2	5.00
0.5	1.25
0.25	0.63

3. The table shows the amount of money a tourist would receive in Egyptian pounds in exchange for different amounts of money in British pounds.

- a Complete the table with the missing values if the exchange rate is 1 British pound to 21.48 Egyptian pounds.

British Pounds, x	Egyptian Pounds, y
100	
250	
	100

- b Write an equation to represent the amount of Egyptian pounds, y , the tourist receives in exchange for x British pounds.

4. Han makes fruit punch following a recipe that uses one 48-ounce can of pineapple juice to one 6-ounce can of frozen lemonade.

- a Complete the table.
- b Write an equation to model how many ounces of pineapple juice, p , are needed for f ounces of frozen lemonade.

Pineapple Juice (oz)	Frozen Lemonade (oz)
48	6
16	
12	
4	

5. Bard buys a half pound of turkey breast for \$4.99 at the local deli.

- a What is the price per pound of the turkey breast?
- b Write an equation to represent the price in dollars of the turkey breast, y , given its weight, x , in pounds.
- c Use your equation to complete the table.

Weight (lb)	0.33		1.82	
Price (\$)		6.75		26.94

6. Noah and Andre examine the table shown representing a proportional relationship. Noah writes the equation $y = 1.5x$ to represent the relationship. Andre writes the equation $y = \frac{2}{3}x$ to represent the relationship.

Who is correct? Explain your thinking.

x	y
12	8
39	26
57	38